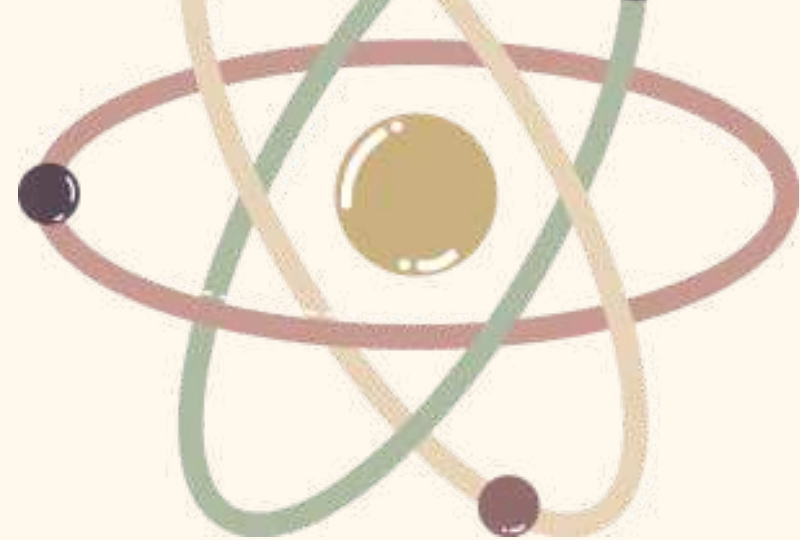
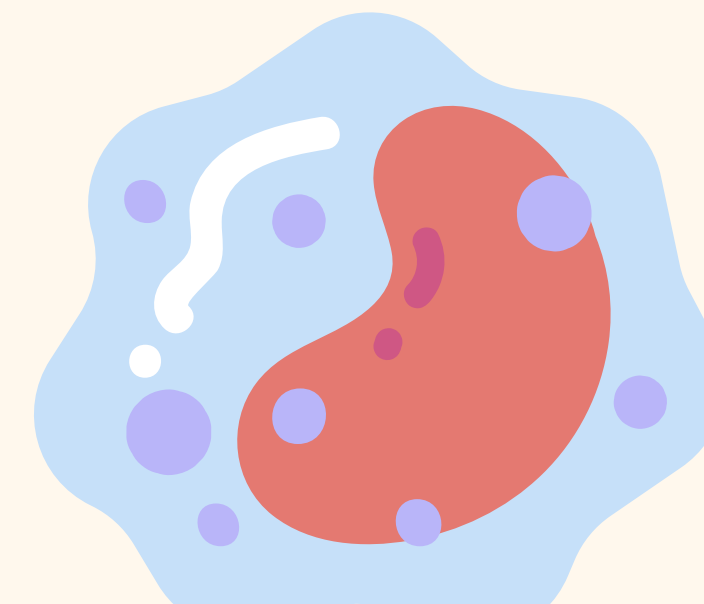
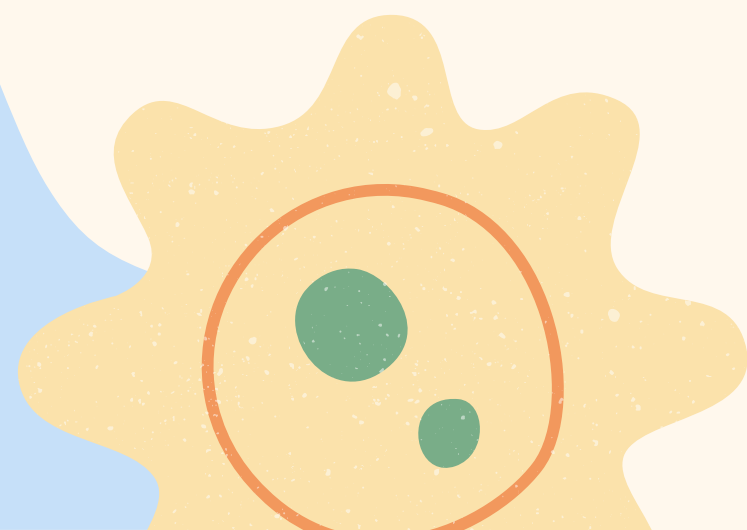
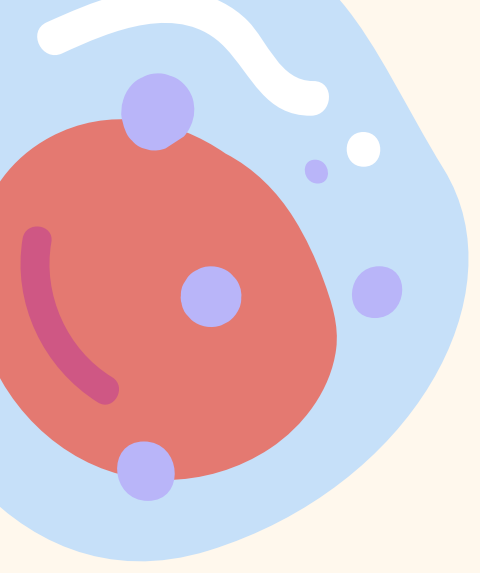




基因密碼解密 —— 從 DNA 到未來糧食安全

Decoding the Genetic Code:
From DNA to the Future of Food Security





Today's Lesson

**what is iGEM/
synthetic
biology?**

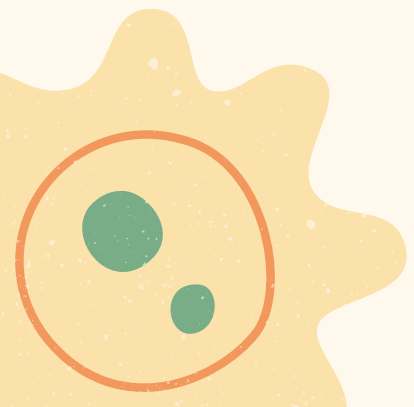
**什麼是iGEM/
合成生物學？**

**Plants living
under stress**

在逆境中生存的植物

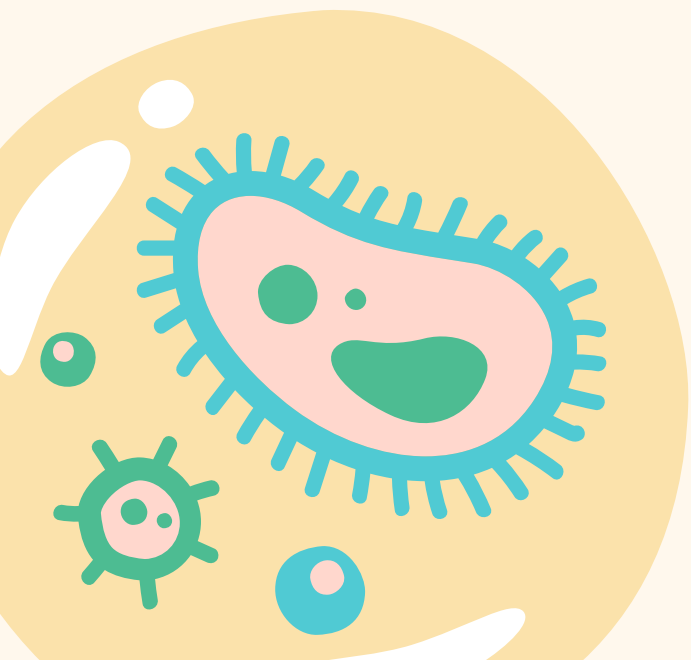
**DNA &
Experiment**

**DNA 遊戲以及
植物逆境模擬實驗**

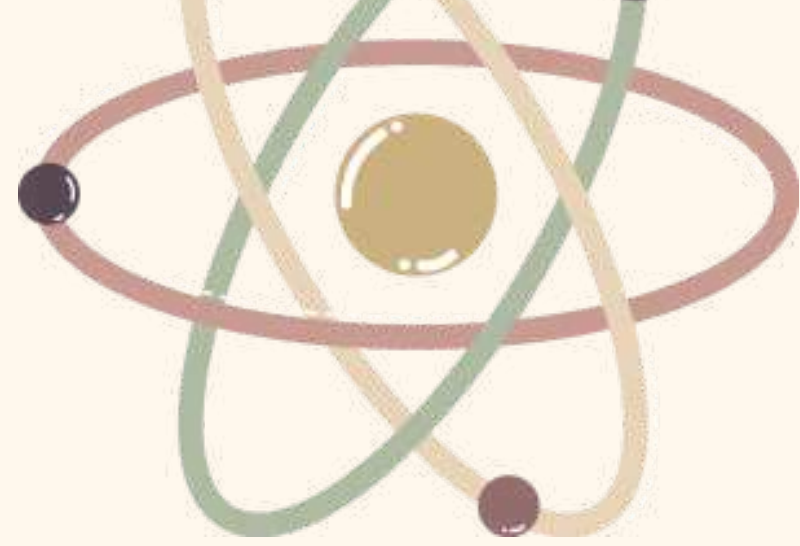




Kahoot!

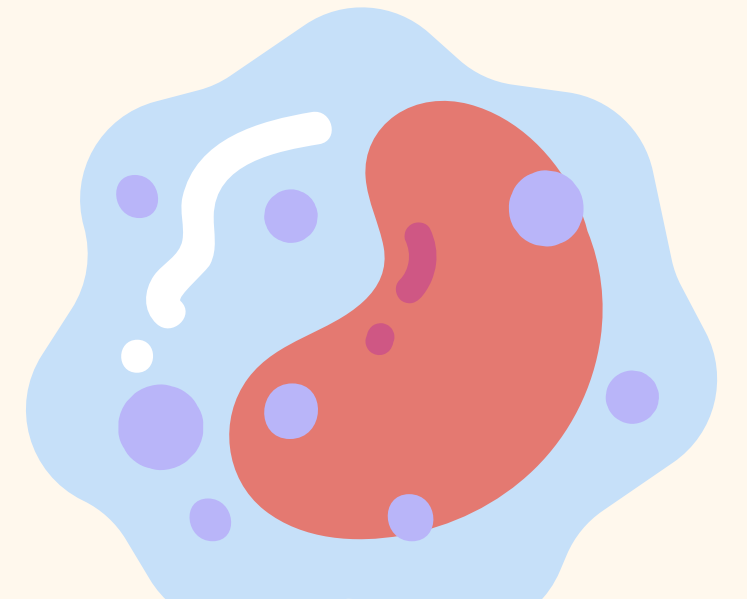
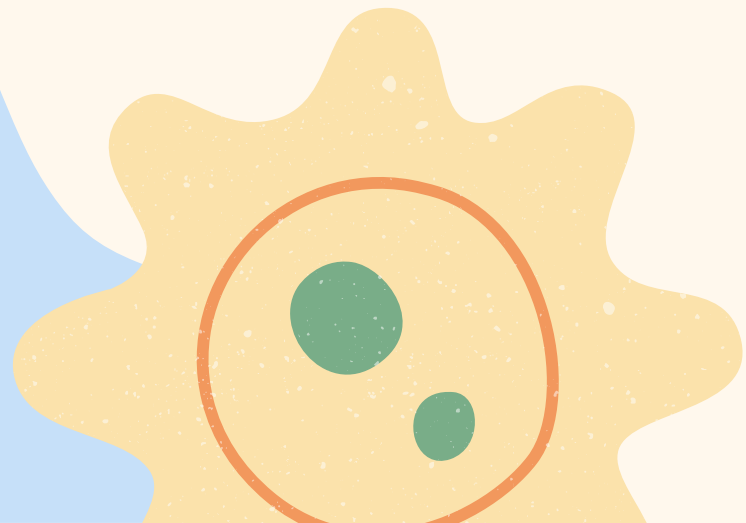


<https://create.kahoot.it/share/8-14-high-school-education-lesson/c47da741-5b16-499d-a208-c248effbec4b>



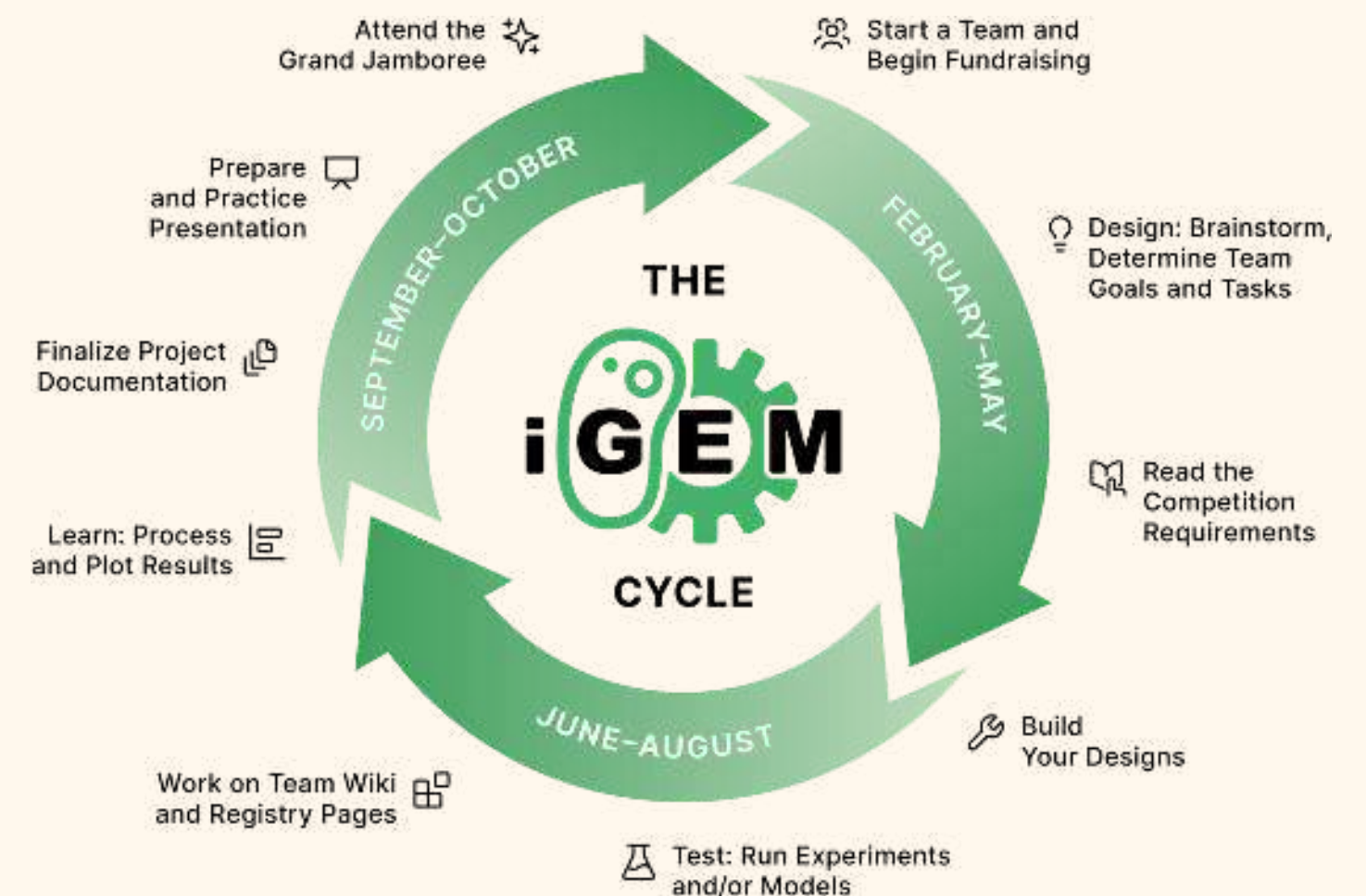
01

Synbio & iGEM



What is iGEM 什麼是iGEM ?

- advancing the field of synthetic biology and building the global synbio workforce through education and competition
- strives to find solutions for a healthier, more resilient and sustainable world
- 透過教育和競賽來推進合成生物學領域的發展
- 致力於為更健康且具備韌性的永續世界尋求解決方案



Synthetic biology use engineered microorganisms to detect, respond to, and solve real-world problems



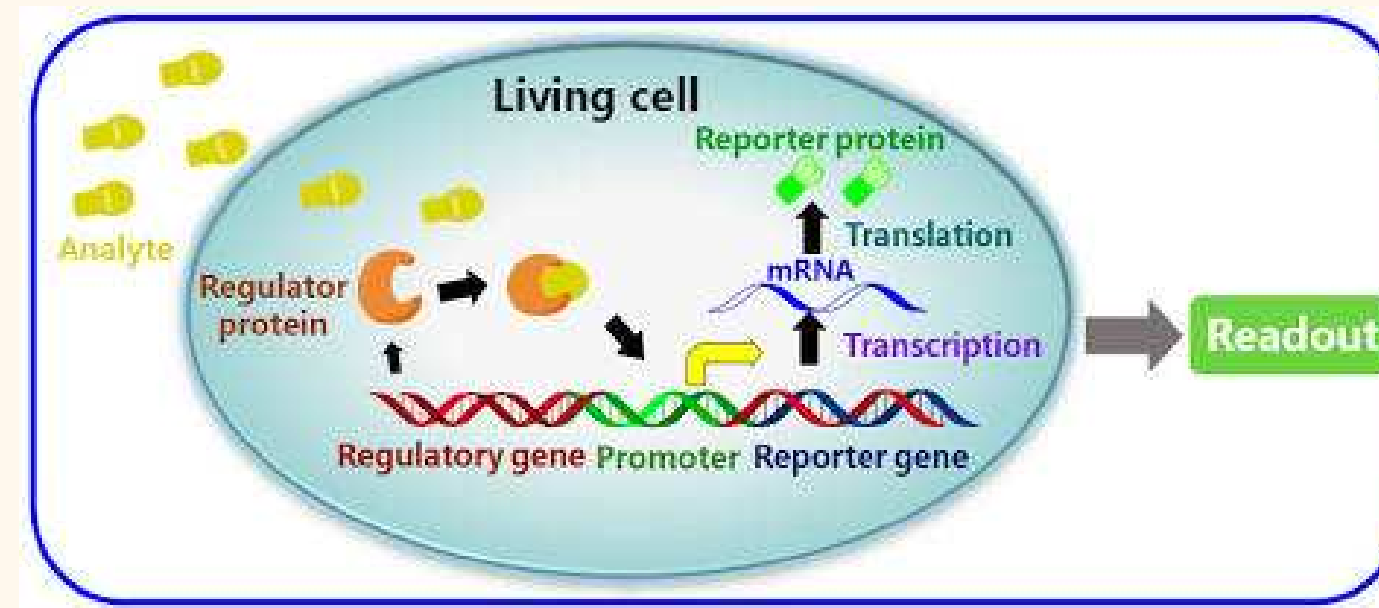
Engineer microorganisms to produce biofuels



Producing medicines using engineered bacteria



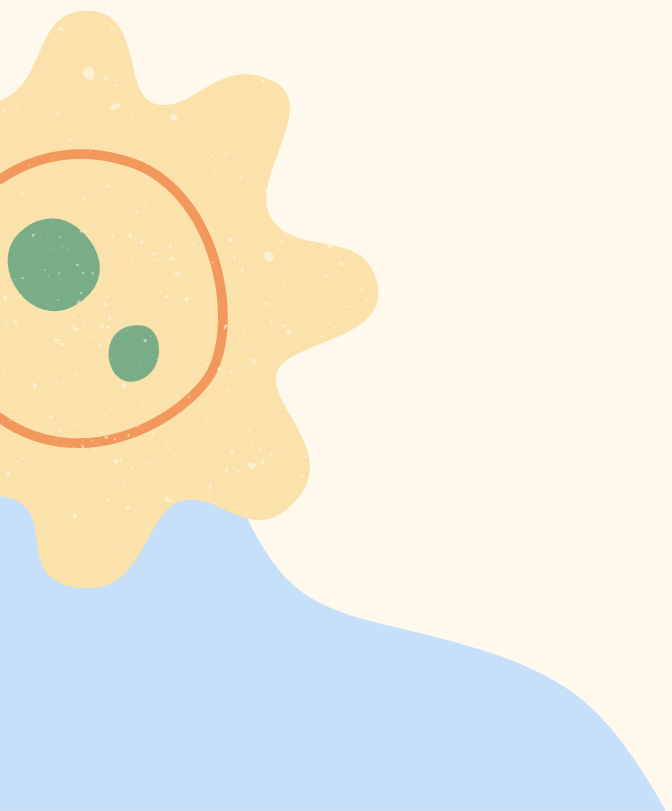
Creating biosensors that detect and remove environmental pollutants





02

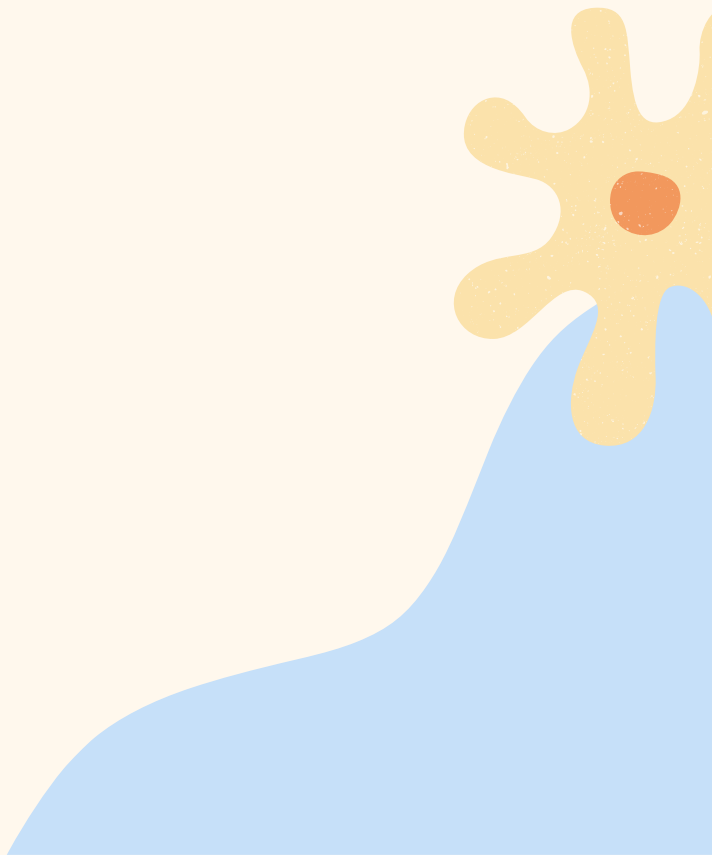
Plant Protectants, Biotic & Abiotic Stress, and Osmosis Experiment





**As the Earth gets warmer and drier,
what will happen to the plants?**

**當地球變得更熱、更乾燥時，
植物會有什麼反應？**



Abiotic vs Biotic Stress 非生物與生物壓力

Two major categories of plant stress and their common examples

Abiotic stress

非生物壓力－來自物理或化學環境



1 Drought 乾旱



2 Salinity 鹽度



3 Extreme temperatures 極端溫度

Biotic stress

生物壓力－來自其他生物體



1 Pathogens 病原體



2 Insects 昆蟲



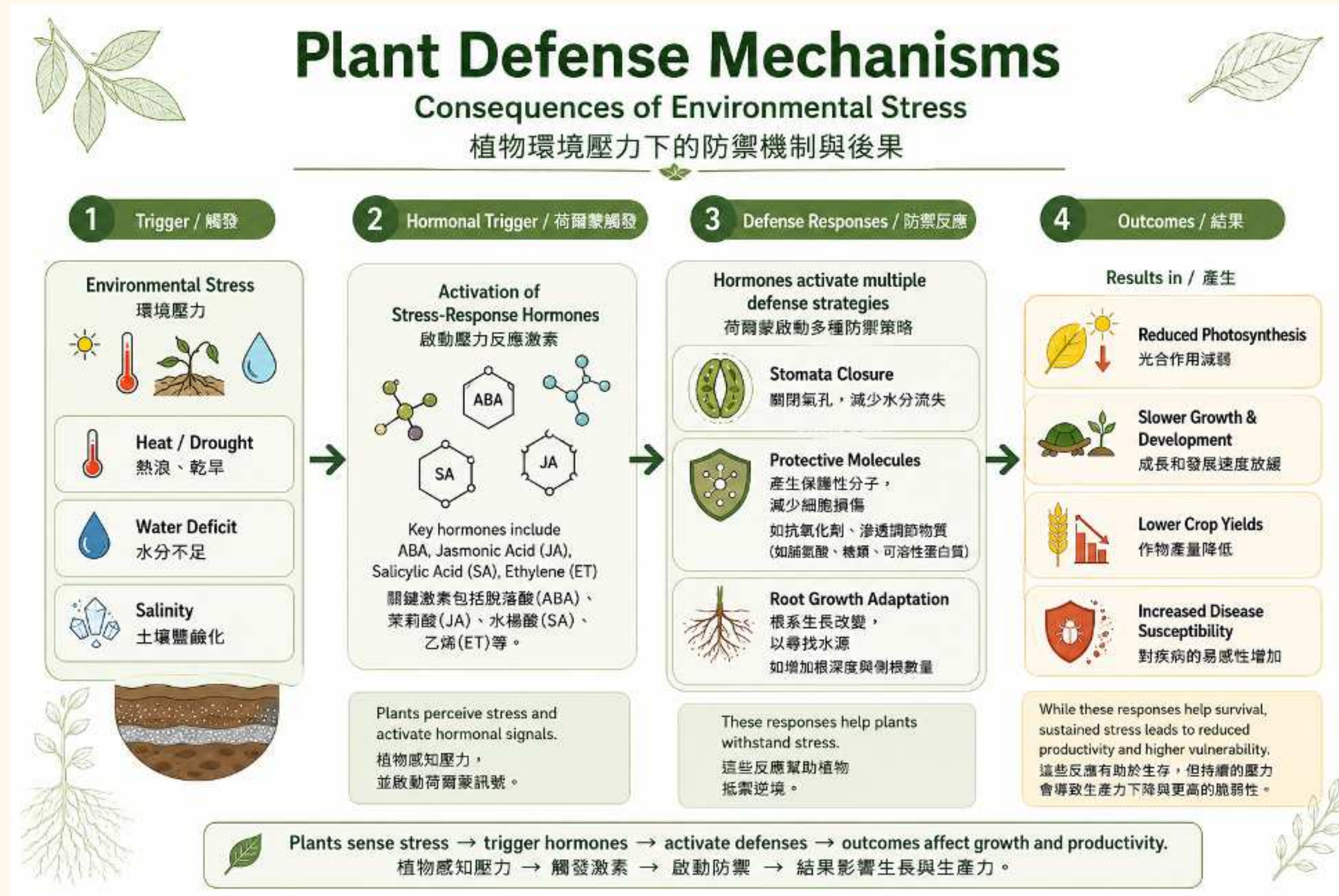
Plant stress can be caused by non-living factors or living organisms.

How does plant stress relate to our daily lives?

植物逆境與我們日常生活有什麼關係？



How does environmental stress affect plant growth, and what are the consequences for food production?

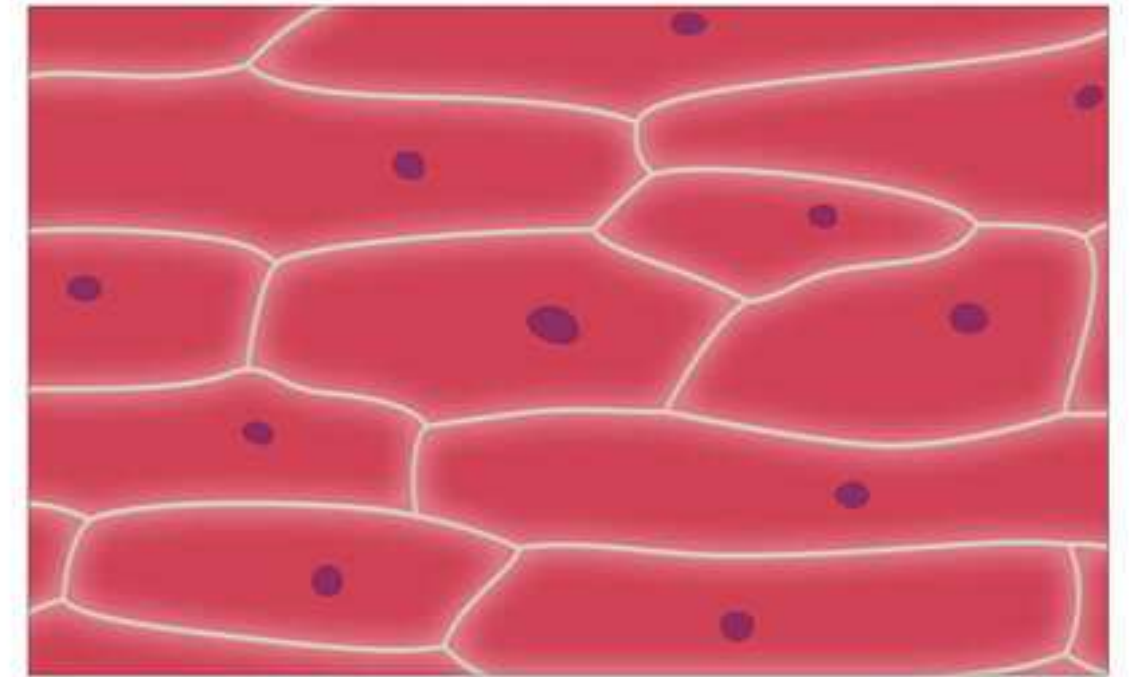


Question:

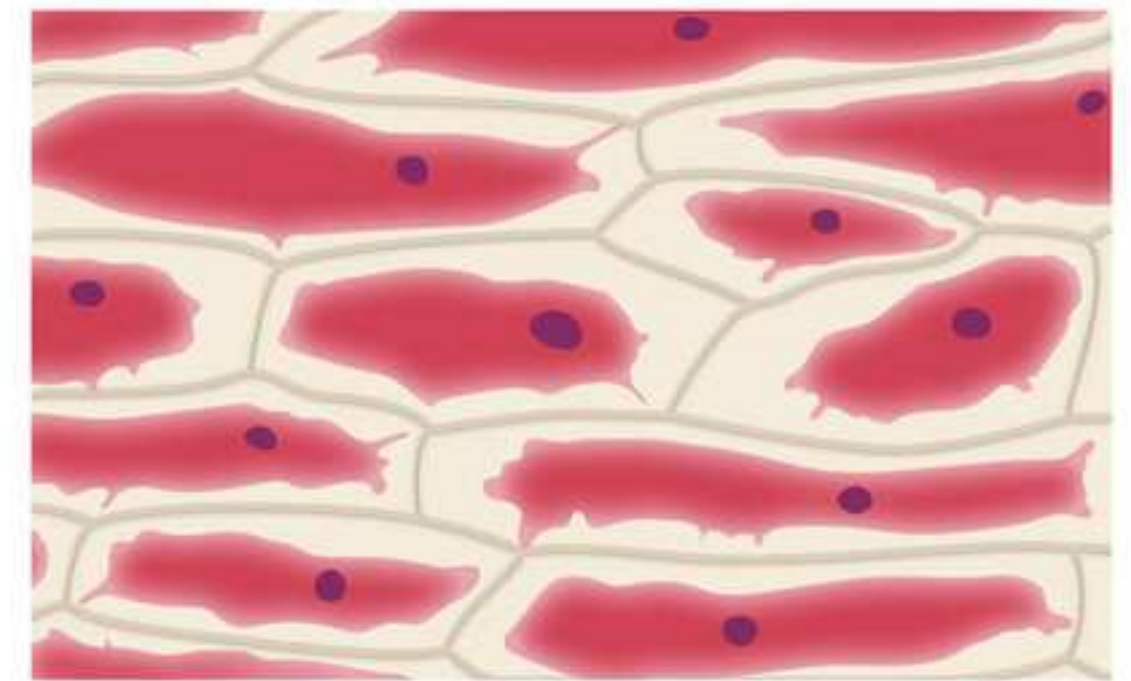
**How does salinity stress affect the
cell wall of onion?**

Experiment - Plasmolysis

- Think about these questions:
 - What will happen when plants are living under salinity stress?
 - How does the onion change in color and texture?
 - How does stress affect how the plant grows?
- What are the consequences?



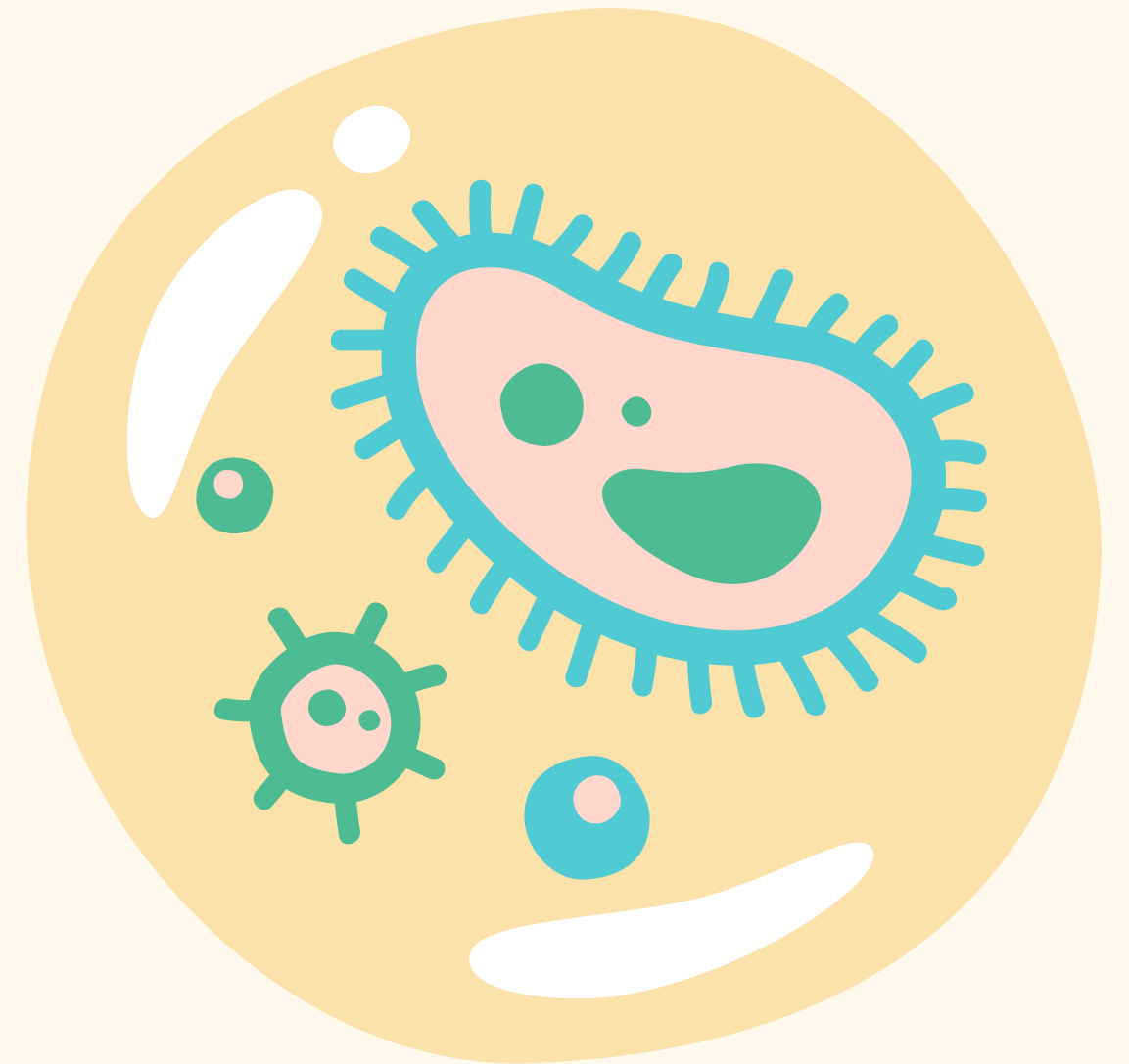
After expose to osmotic stress



Experiment - Plasmolysis

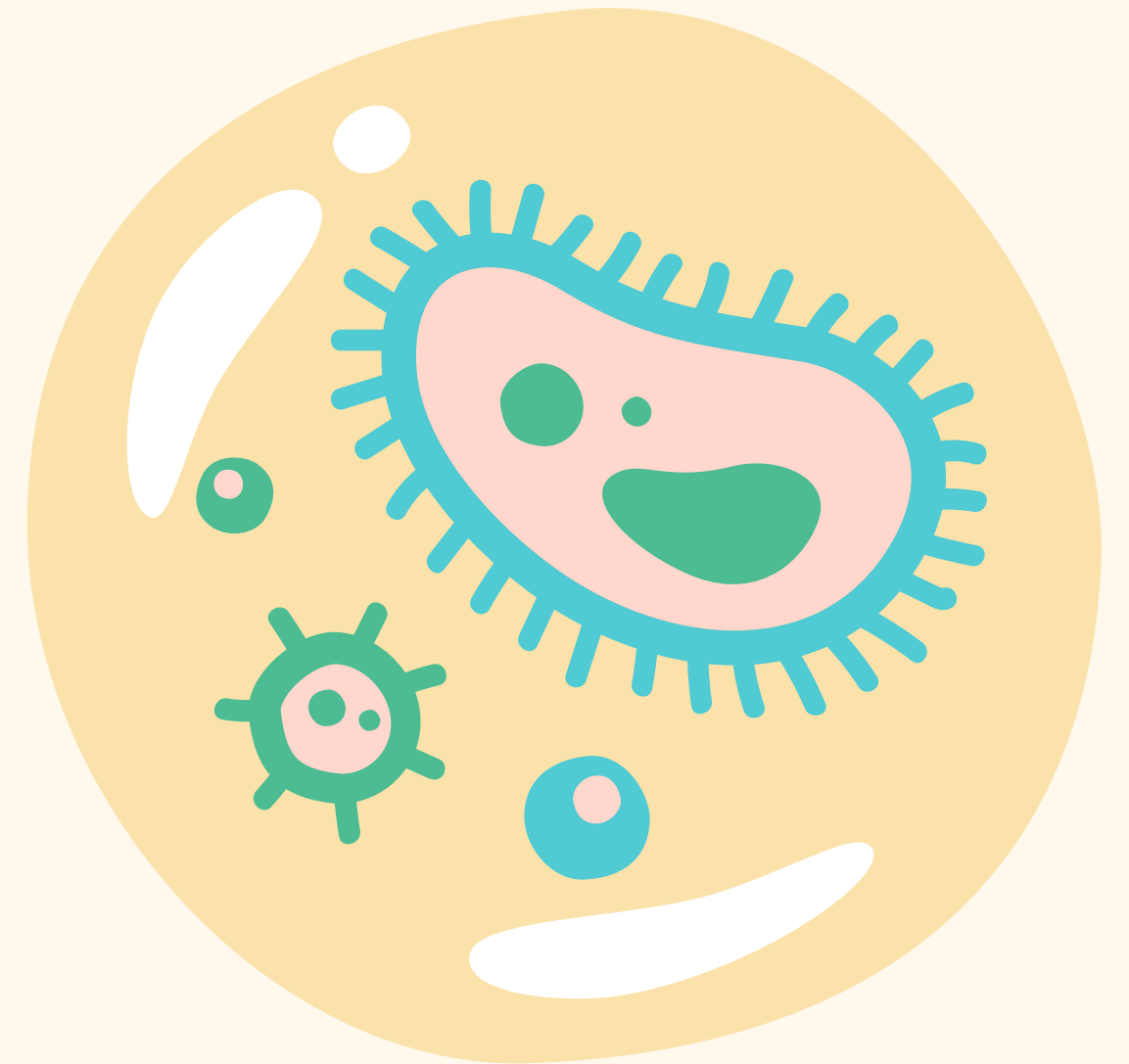
- materials

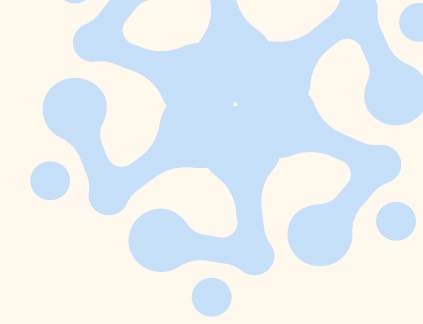
- onion
- tweezers
- dropper
- salt solution (3M and 6M) in centrifuge tubes
- absorbent paper (吸水紙)
- microscope slide
- cover glass
- microscope



Experiment - Plasmolysis

- In this experiment, we will see:
 - Changes in color after soaking in different solutions
 - Different effects of plasmolysis under different concentrations of salt





Procedure - Plasmolysis

Step 1:

Peel a thin layer of onion with tweezers



Step 2:

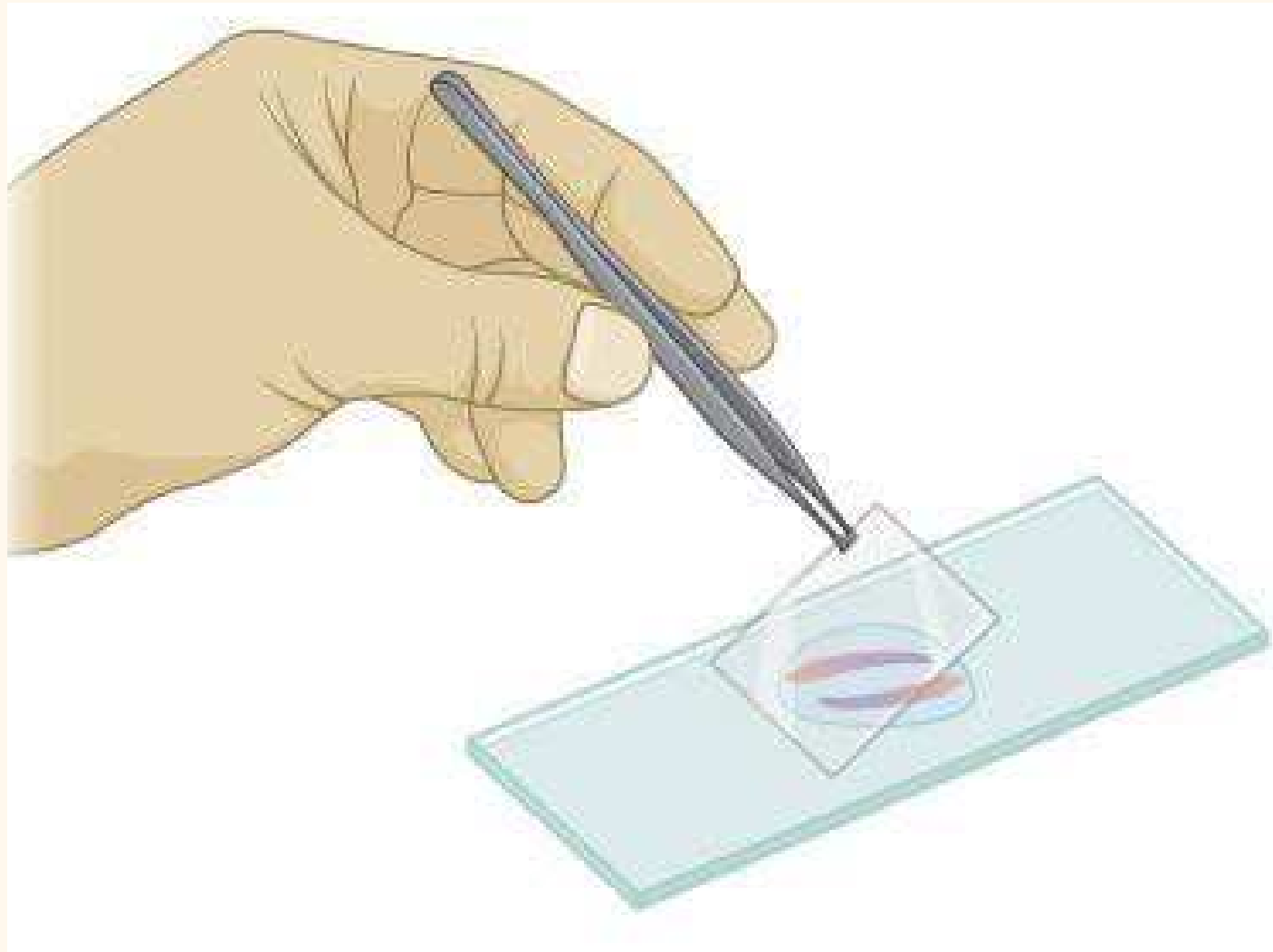
**Put the onion on the microscope glass,
use the dropper to drop distilled water on it**



Step 3:

Put the coverslip on the onion

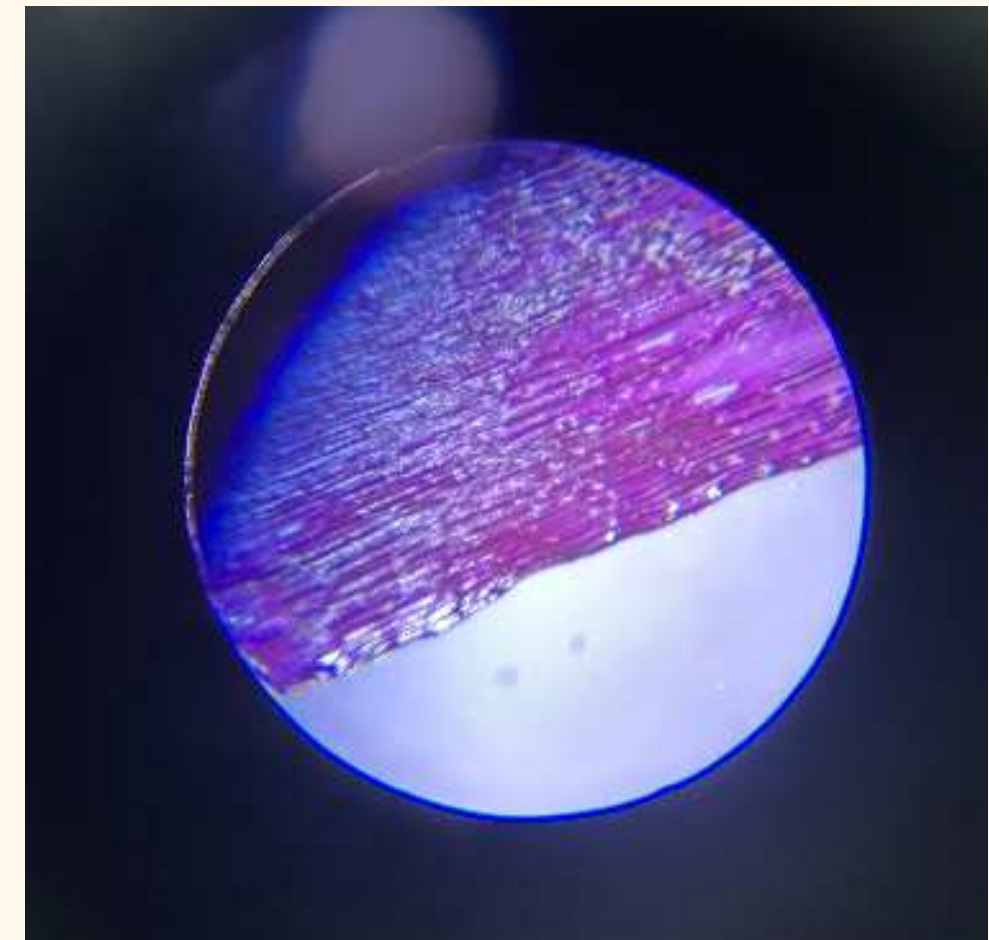
Use the microscope to observe the onion cell.



Step 4:

Take a photo of the onion cell through the microscope . (Before plasmolysis)

****Remember to upload the photo to the drive
by our team member****



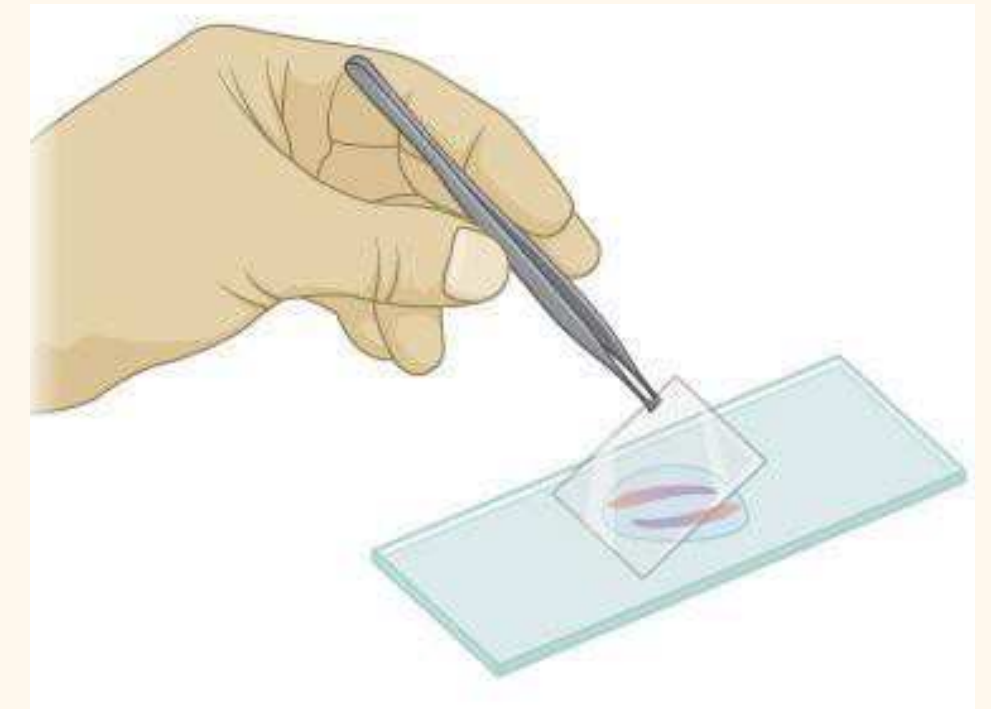
Step 5:

Take off the coverslip.



Step 6:

Open the Salt solution tube, use a new dropper to drop the salt solution on the onion on the microscope slide and cover it with coverslip.



Step 7:

**Wait for 15 minutes to let the onion cell
immerse in the salt solution.**



Step 8:

**Take a photo of the onion cell through the microscope
and upload to the drive. (After plasmolysis)**



*Notes

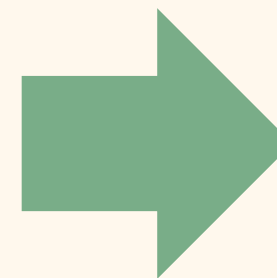
Problem: The onion epidermis may float on top of the solution

→ Prevents the epidermis from not fully absorbing the solution

How to solve: Use the tweezers to turn over the onion epidermis



Floating on the top



Immerse into the solution





The cells have been damaged, so the pigment leaked.

→ Be careful when you peel off the onion epidermis

Video of how heat stress affects photosynthesis:





GEMS Taiwan

We are GEMS-Taiwan, a high school team competing in the Biomanufacturing track. Over the past several months, we have been developing, testing, and refining our project before the final competition this November.



GEMS Academy 2026

Our Project:



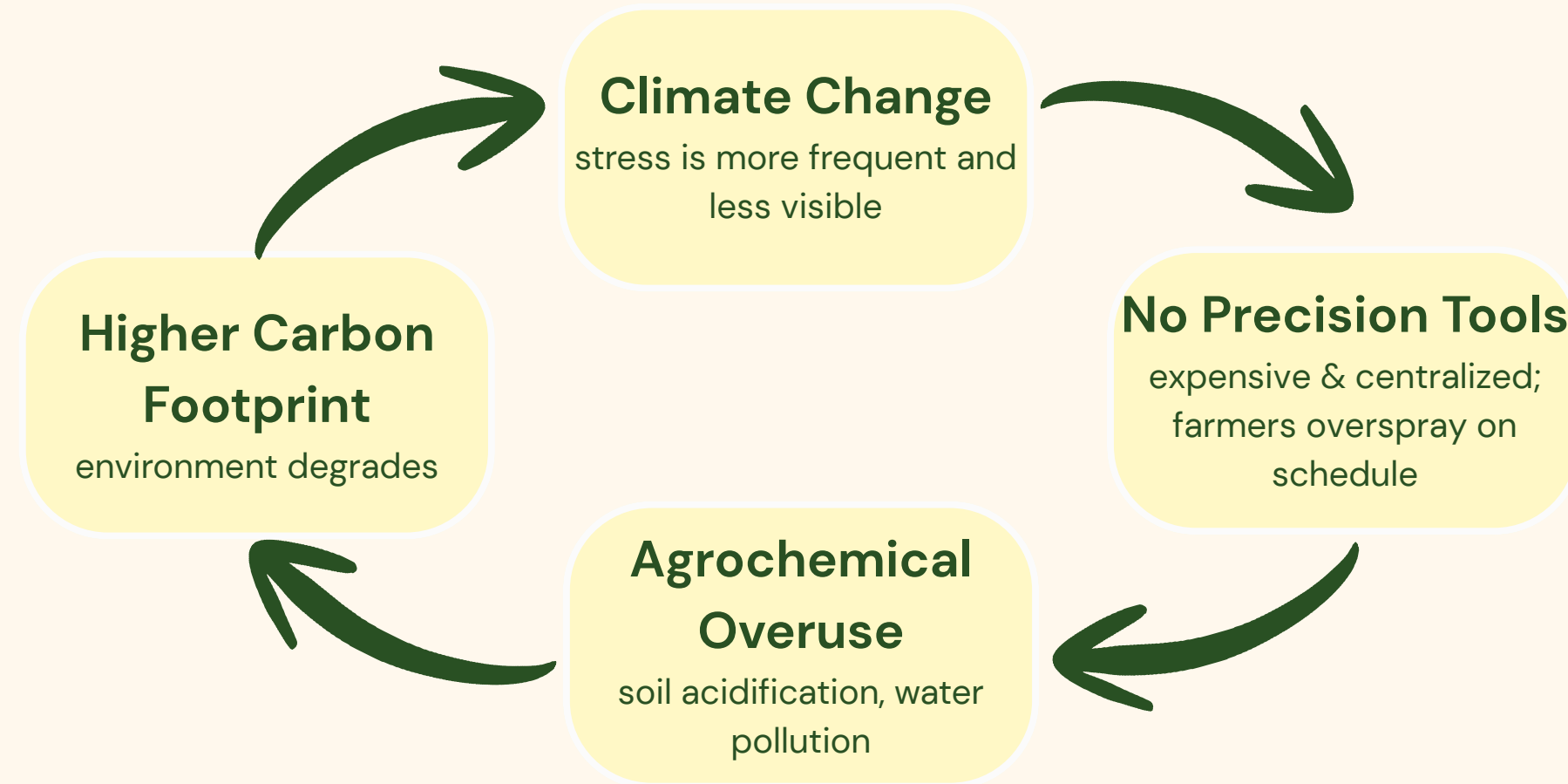
ReLeaf

A Stress-Responsive Optogenetic Bioreactor for On-site Precision Plant Protection



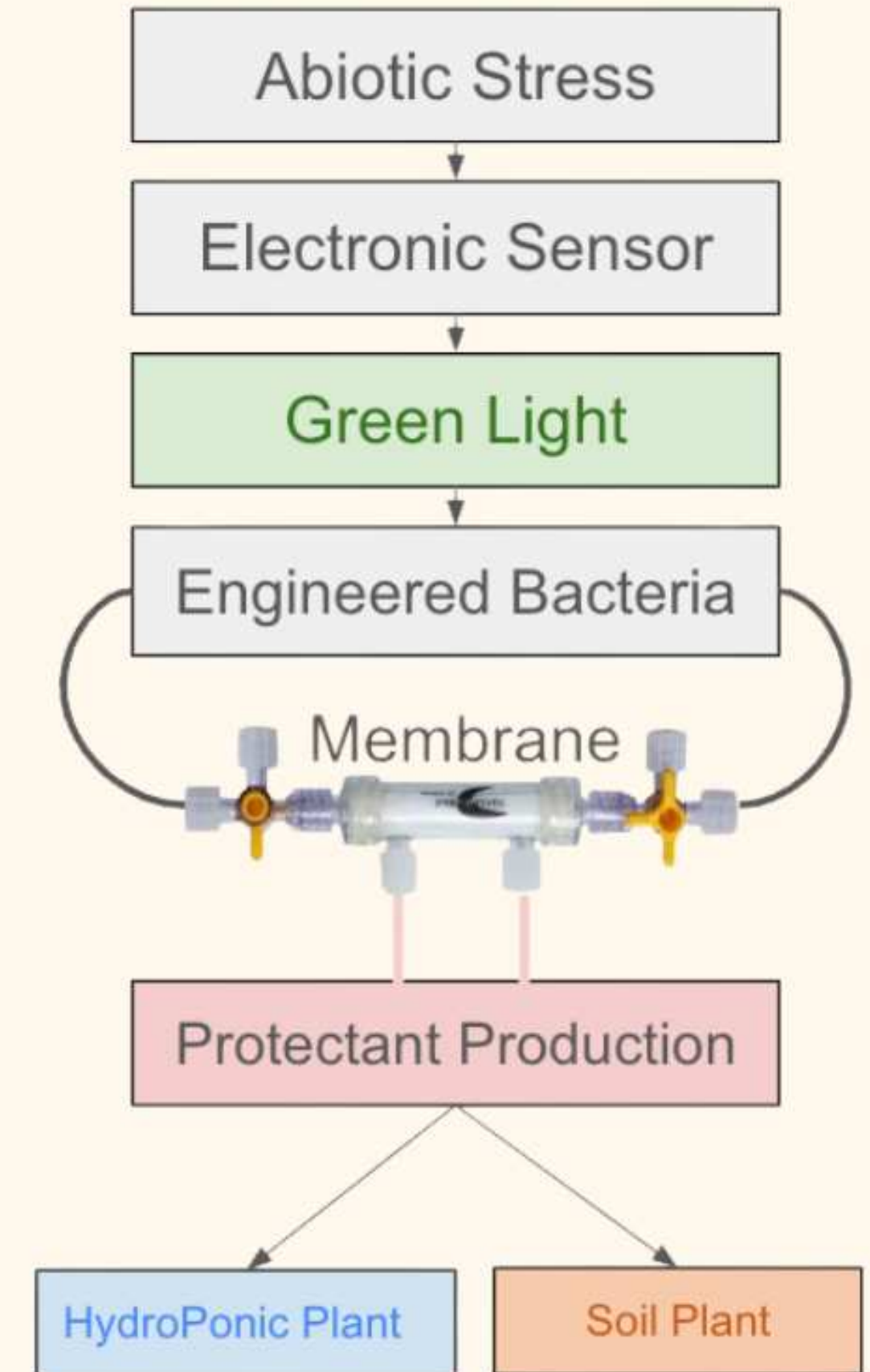
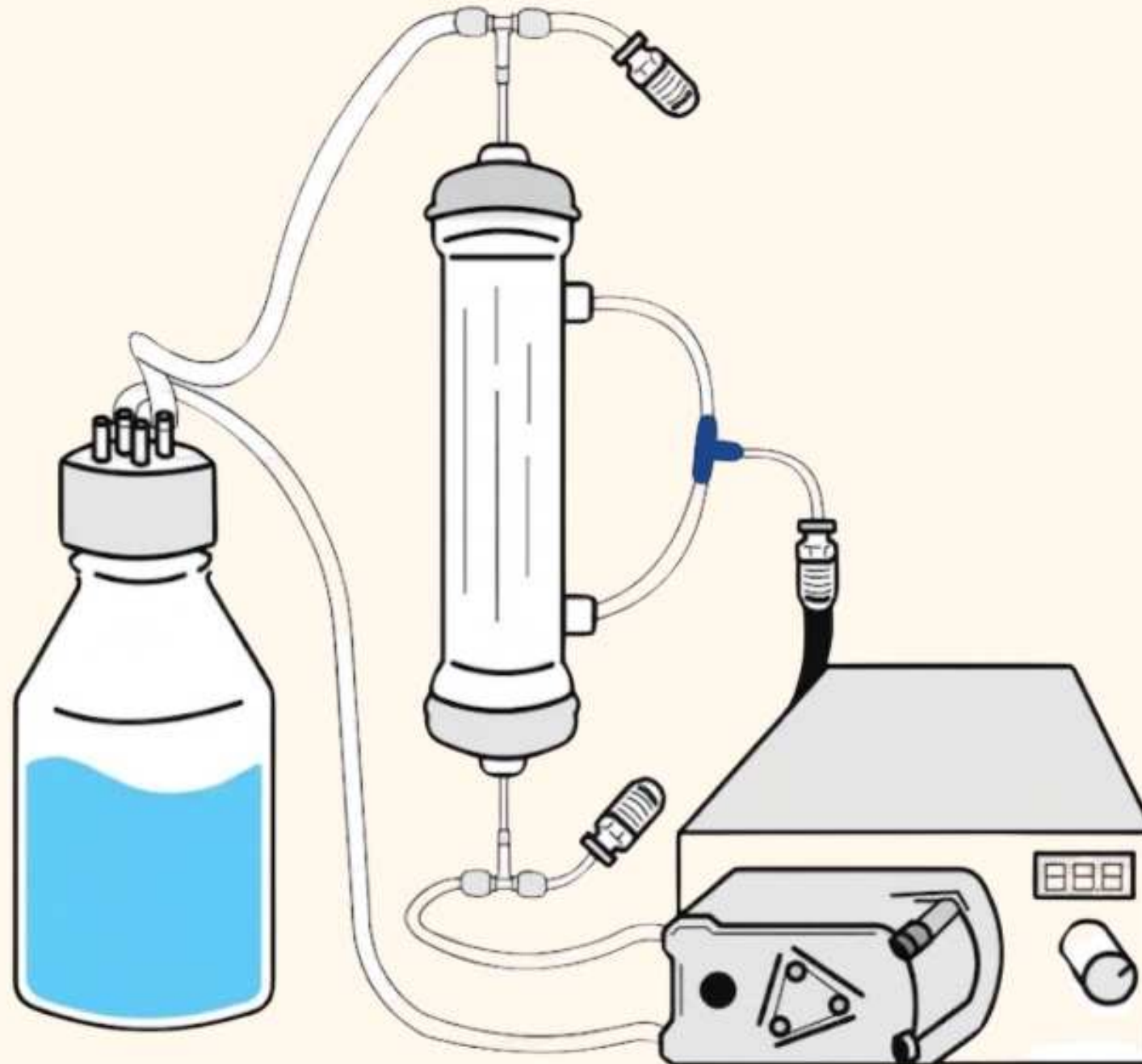
The Problem

The Vicious Cycle



So we asked a simple question: What if protection could be produced directly in the field, with the right amount, exactly when it's needed?

Our Bioreactor Solution



Application Scenario



Bioreactor

Releases proteccants on-site



Method 1: Hydroponics

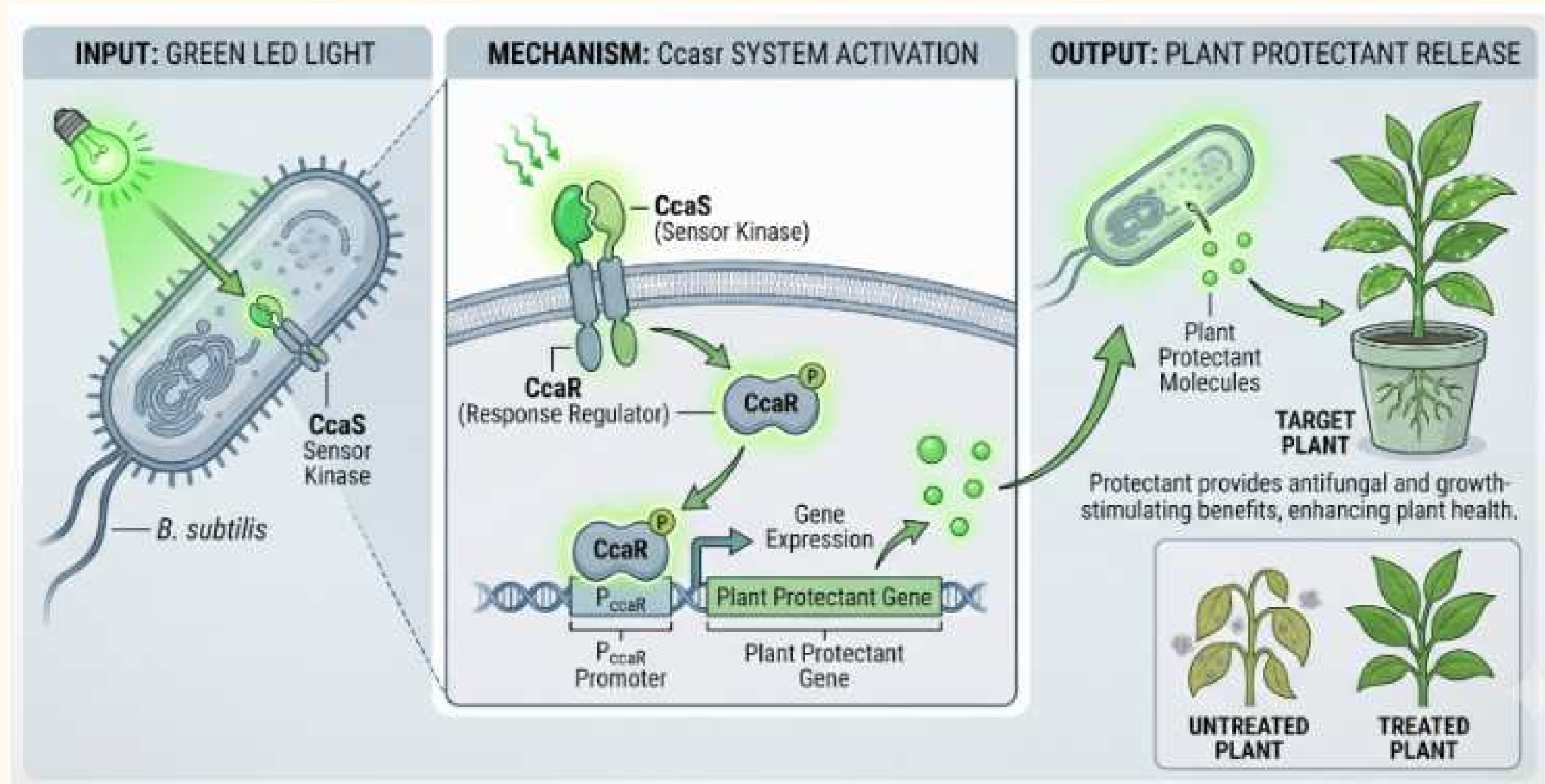
Microbial solution is added into the nutrient water system.



Method 2: Soil Irrigation

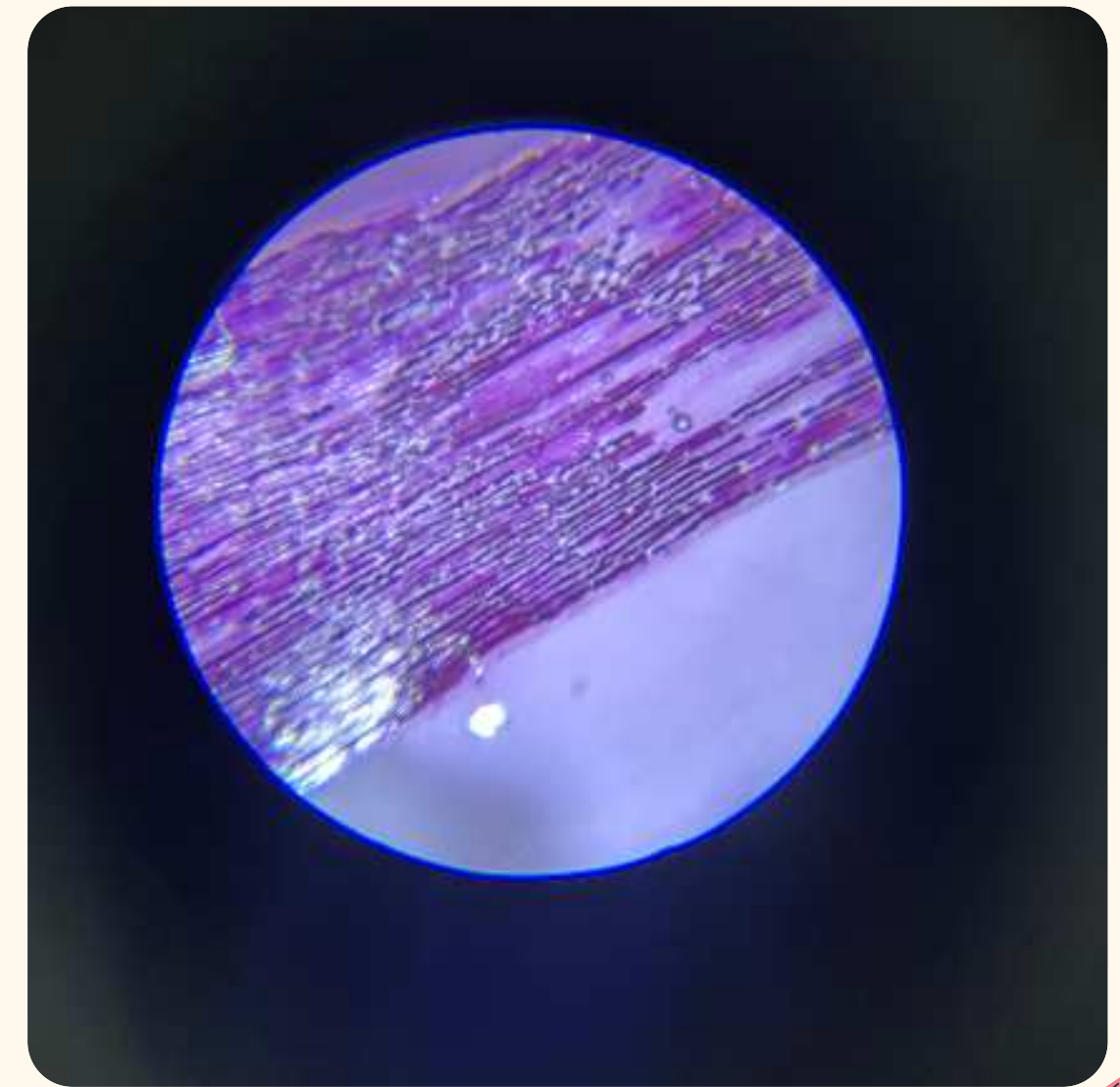
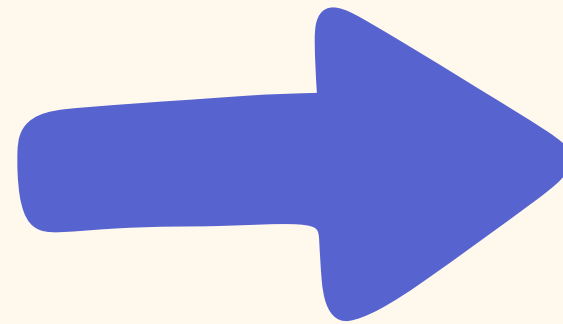
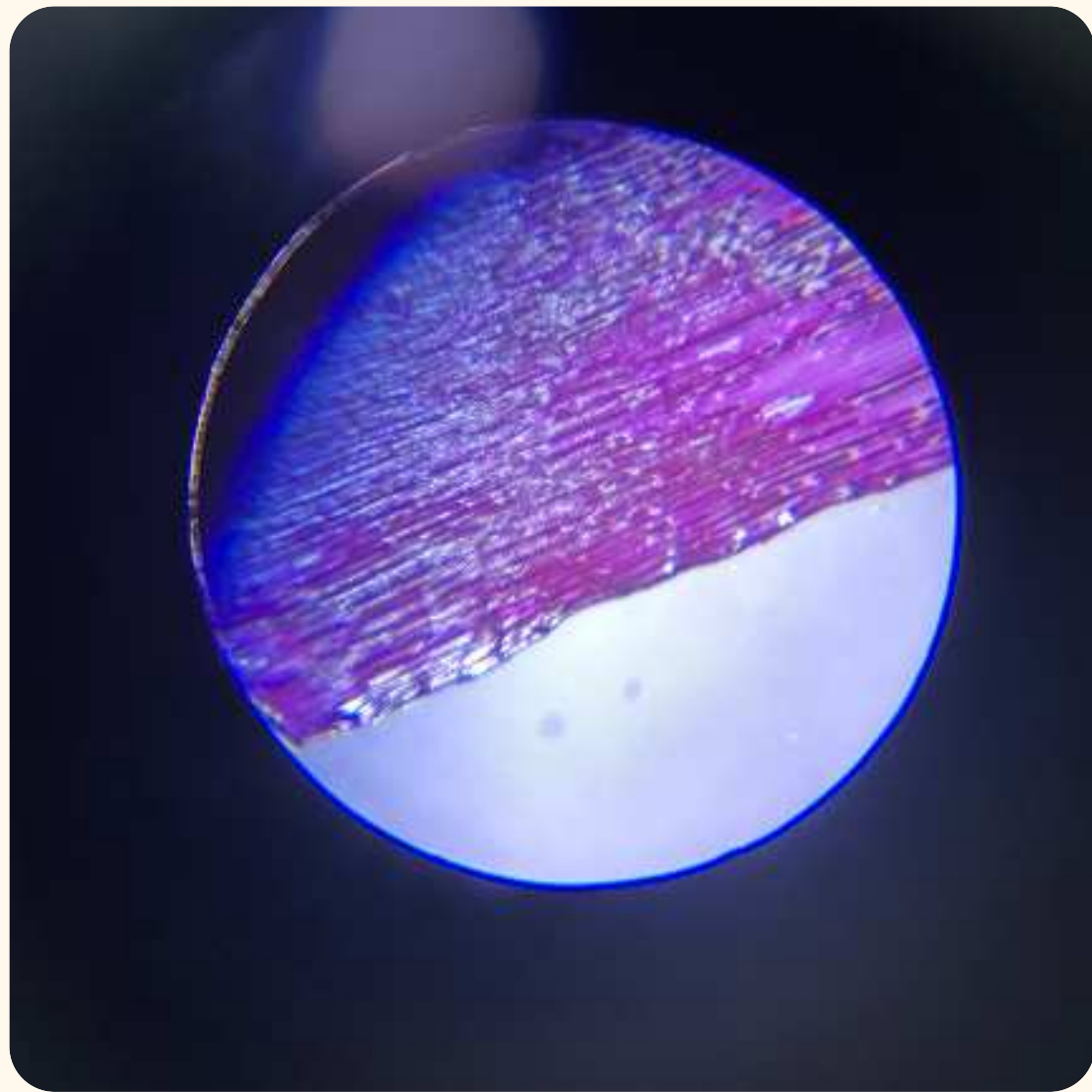
Microbial solution is delivered through soil-level sprinklers.

Synbio Mechanism



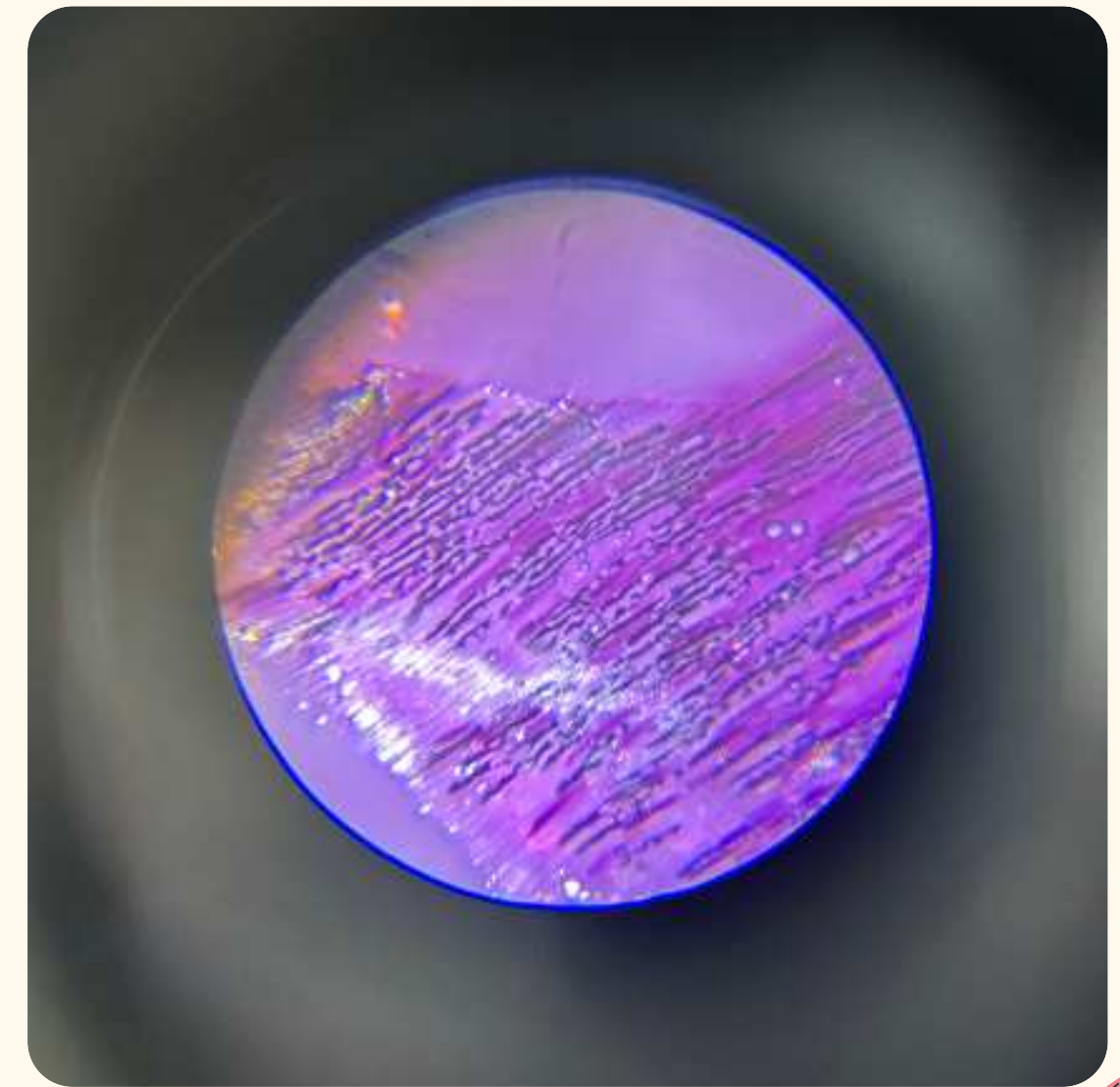
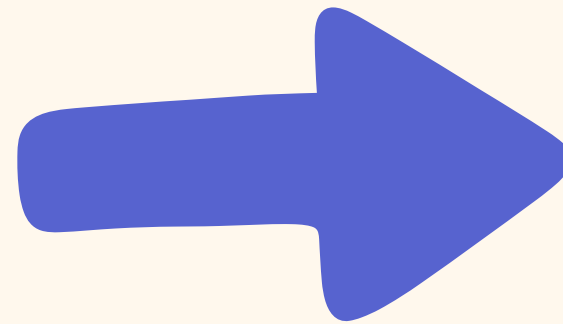
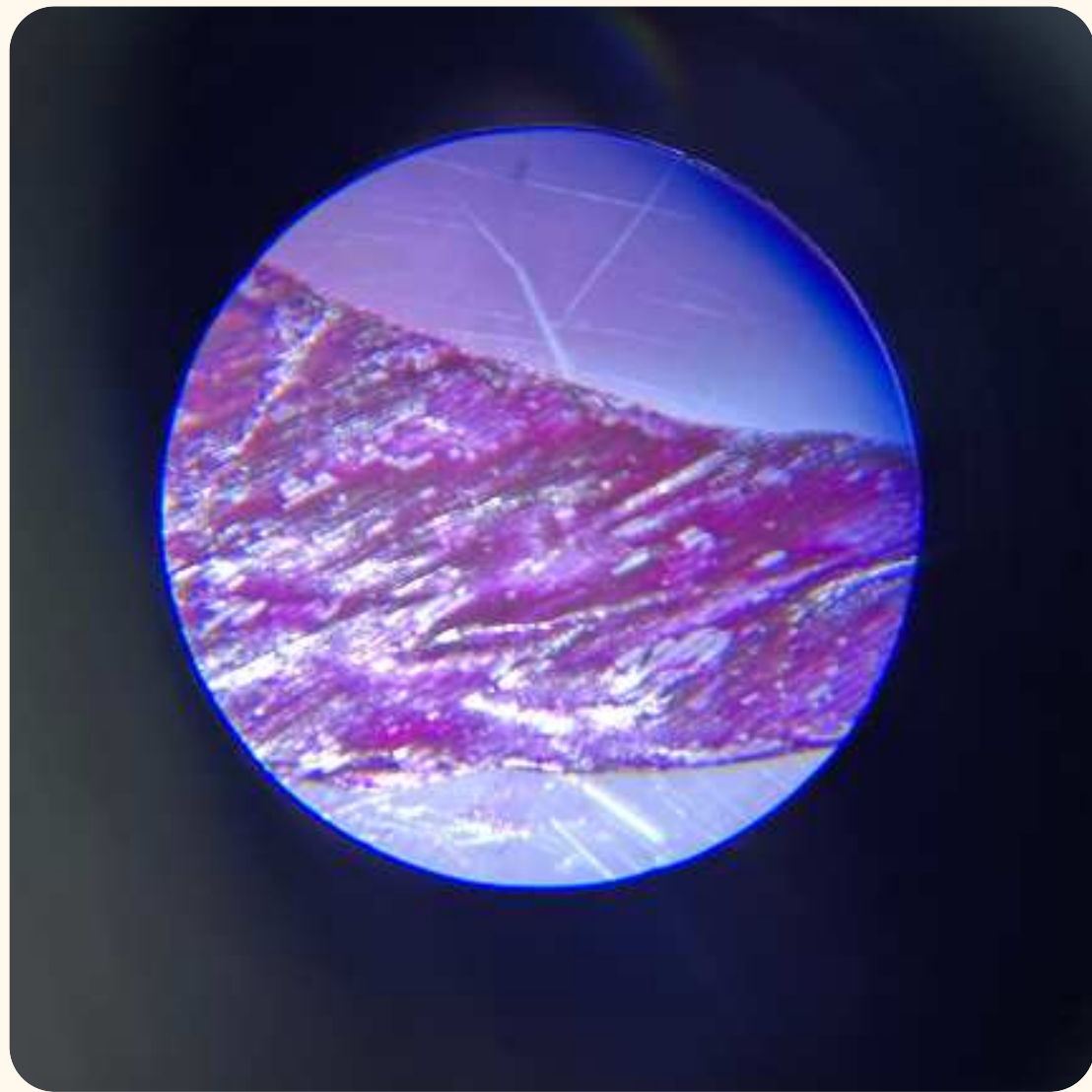
Experimental result - Plasmolysis

6M salt solution



Experimental result - Plasmolysis

3M salt solution

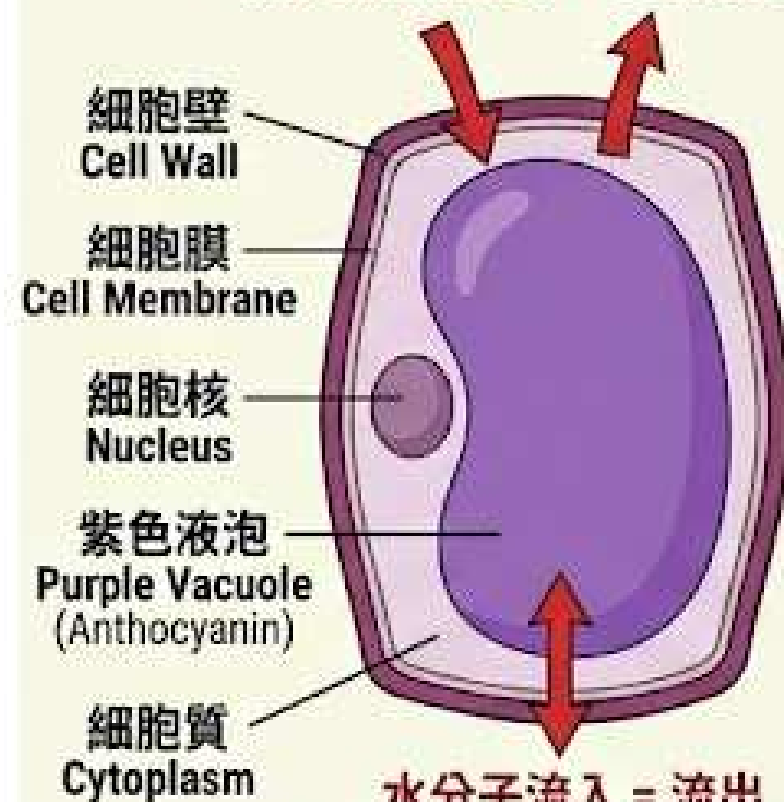


雙語圖解：洋蔥細胞質壁分離與復原實驗

BILINGUAL DIAGRAM: ONION CELL PLASMOLYSIS & DEPLASMOLYSIS EXPERIMENT

1. INITIAL STATE (HYPOTONIC/ISOTONIC) 初始狀態

水分子流入 = 流出
WATER INFLOW = OUTFLOW

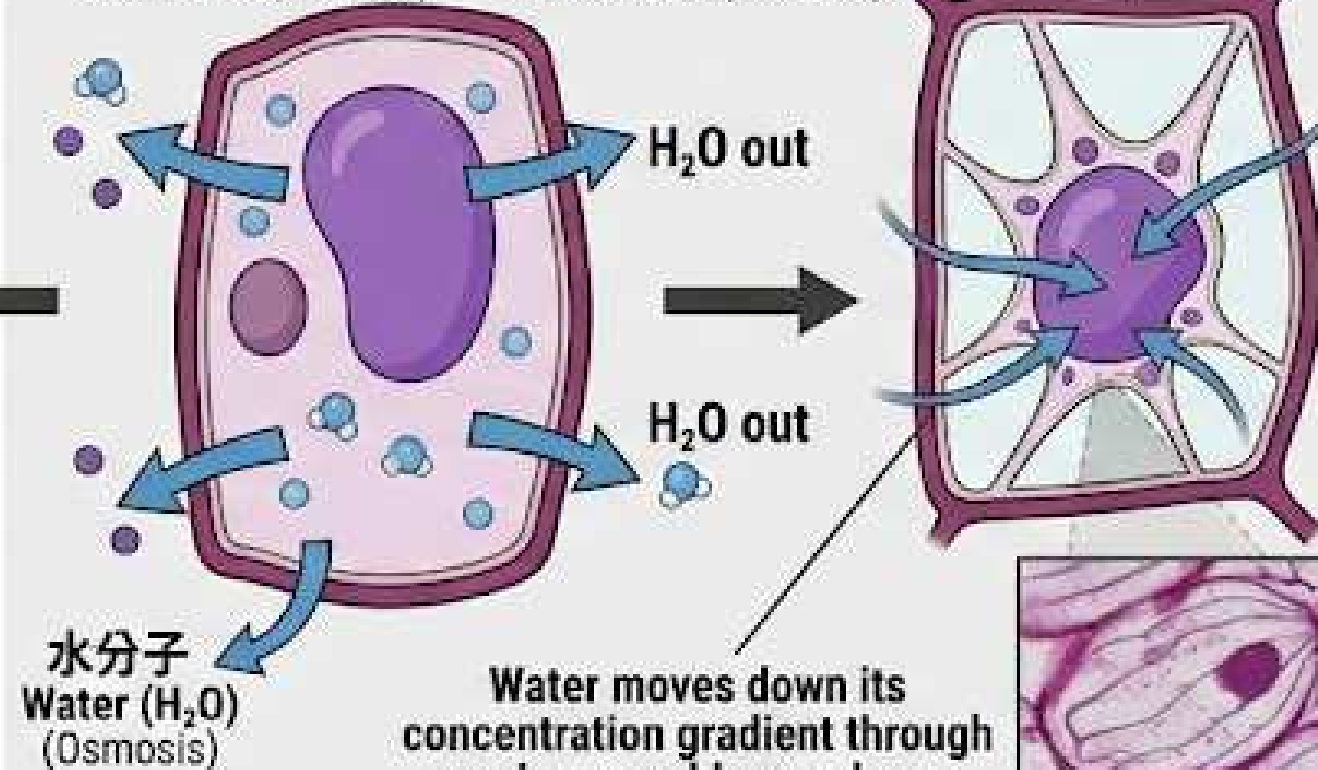


水分子流入 = 流出
WATER INFLOW = OUTFLOW
漲飽狀態
Turgid State: Vacuole full, membrane pressed against cell wall

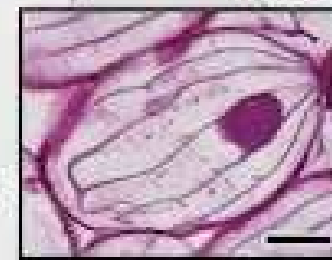
2. OSMOSIS & SHRINKAGE (HYPERTONIC SOLUTION) 滲透與收縮

OSMOSIS PULLS WATER OUT
滲透作用將水分抽離

Low-solute side (Inside) → High-solute side
低溶質側(細胞內) → 高溶質側(鹽溶液)



Water moves down its concentration gradient through semi-permeable membrane
水分沿濃度梯度穿過半透膜



顯微鏡實測圖
Micrograph Observation

3. PLASMOLYSIS & PIGMENT CONCENTRATION 質壁分離與色素濃縮

SAME ANTHOCYANIN VOLUME, SMALLER SPACE
同樣的花青素體積，更小的空間
Pigment squeezed into shrunken protoplast

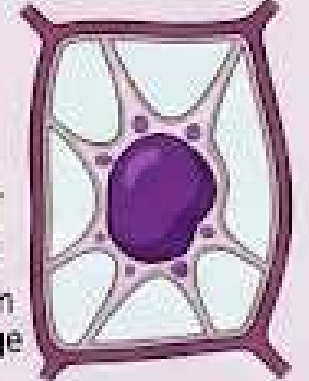
SPREAD PIGMENT
色素分散



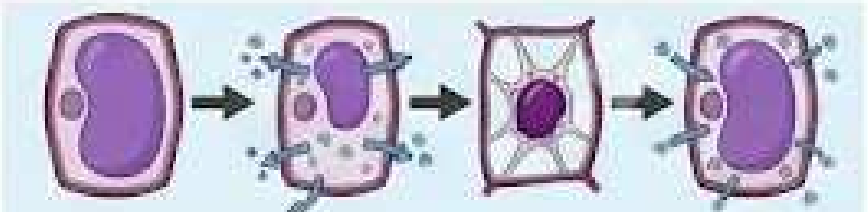
SPREAD PIGMENT
色素分散

COLOR LOOKS DENSER & MORE INTENSE
(no new pigment)
顏色顯得更密集和強烈 (沒有產生新色素)

CONCENTRATED
色素濃縮



CONCENTRATED
色素濃縮

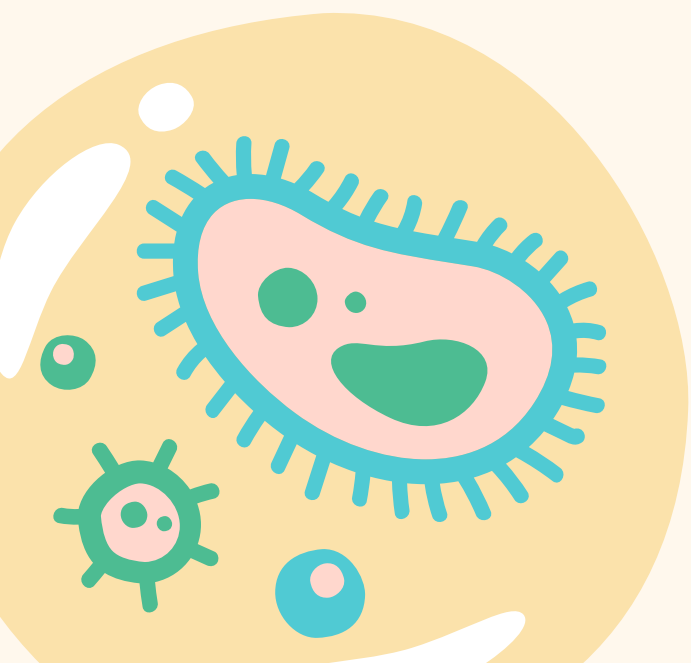


I. → II. → III.

I. → II. → III.

03

DNA & Mutation

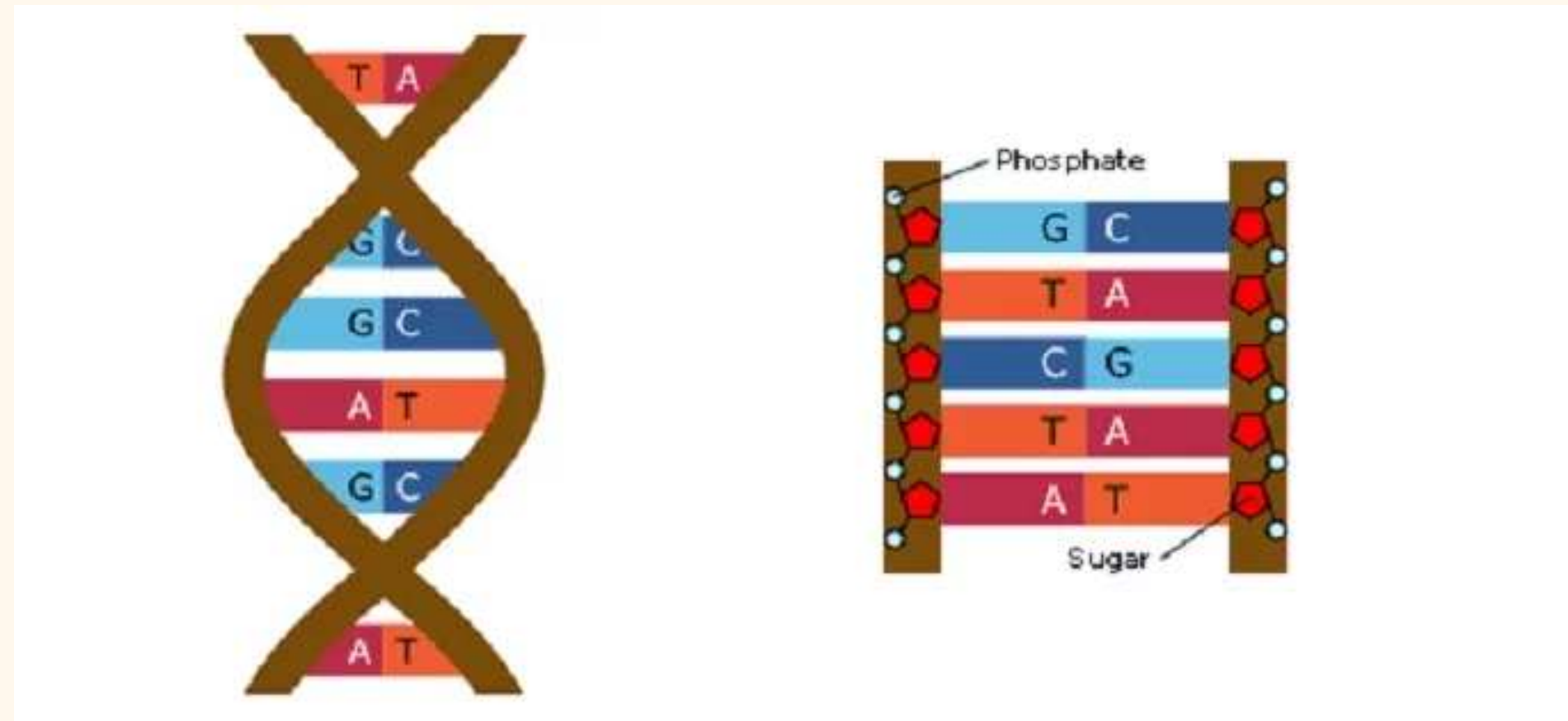
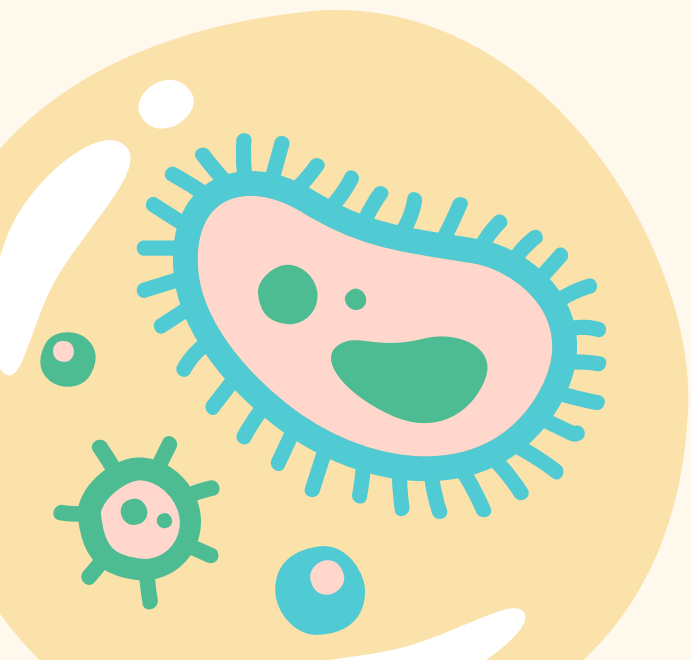




DNA

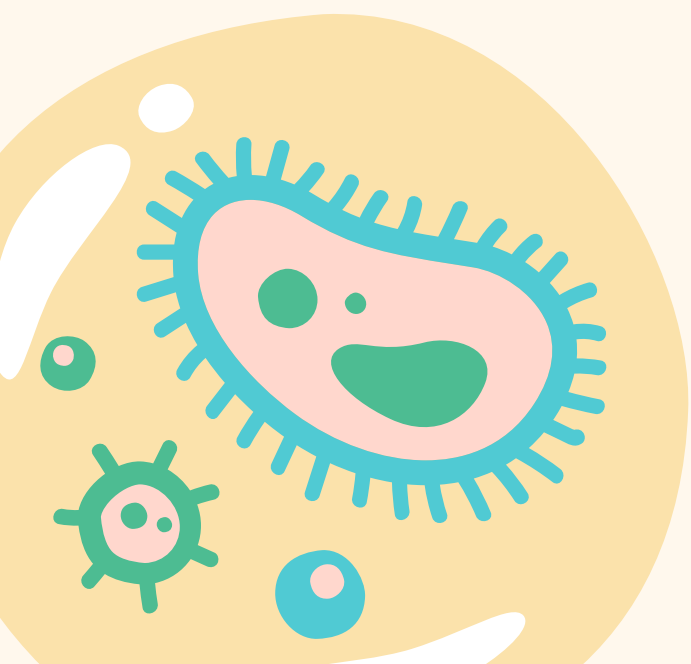
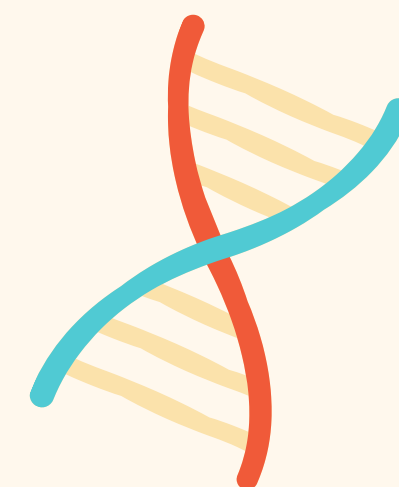
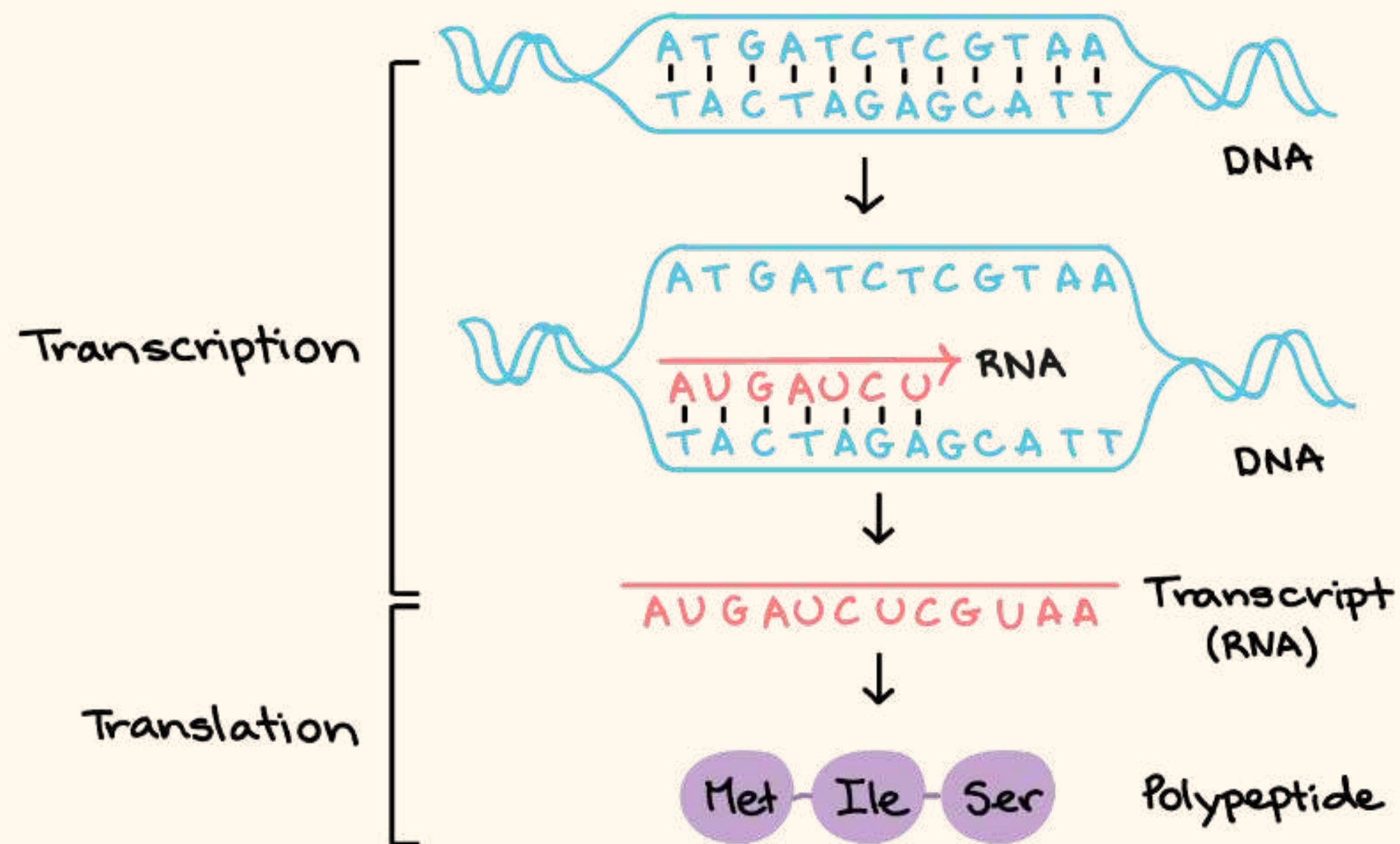


- Code: A,C,T,G
- Rule: adenine (A) always pairs with thymine (T),
and guanine (G) always pairs with cytosine (C)





DNA-transcription and translation

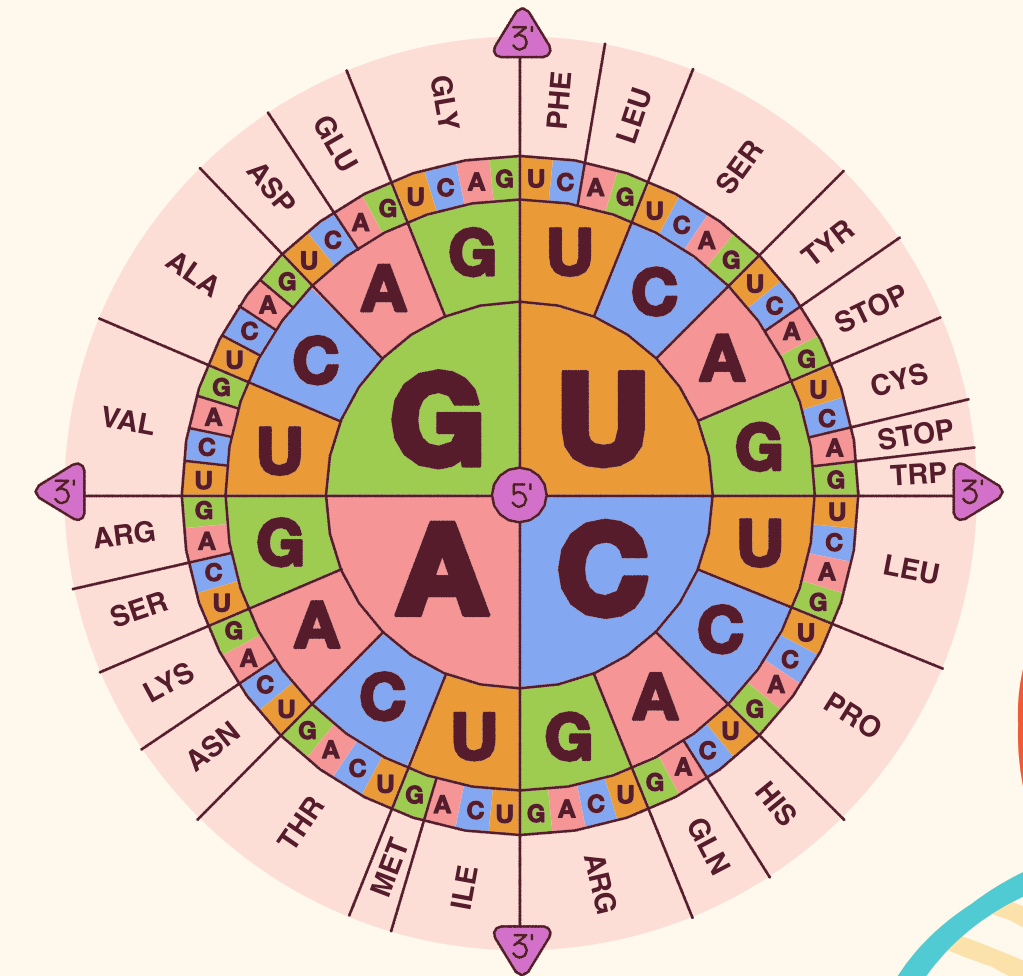




DNA - Amino Acid



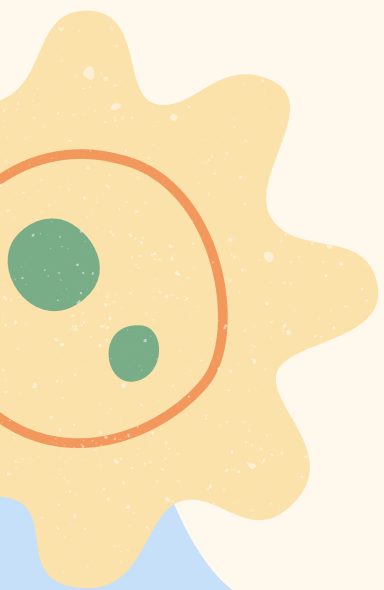
- Molecules that combine to form proteins
 - There are 20 different types of amino acids used by almost all living things.
- During Translation, the Ribosome strings amino acids together in a specific order based on the mRNA code. This chain is called a Polypeptide.





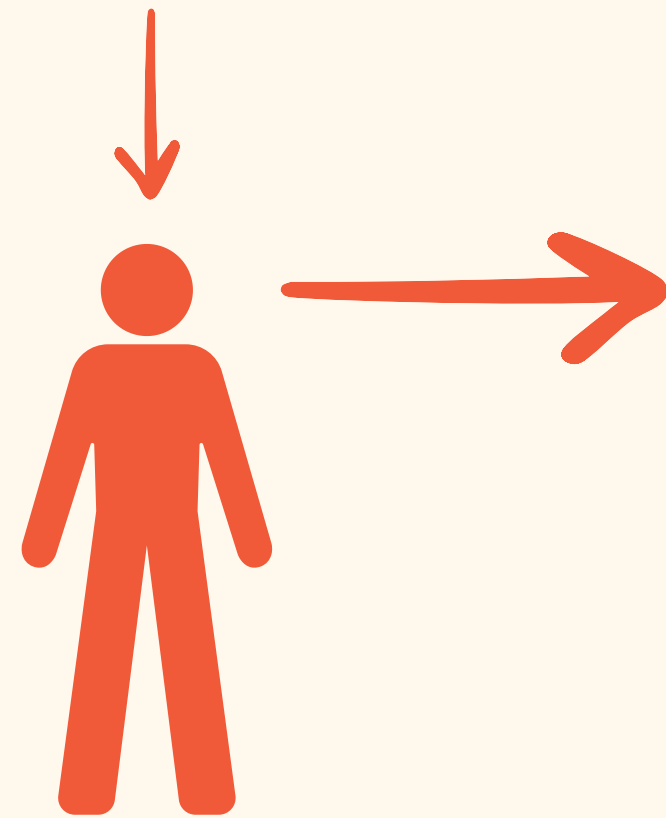
04

Mutation Game



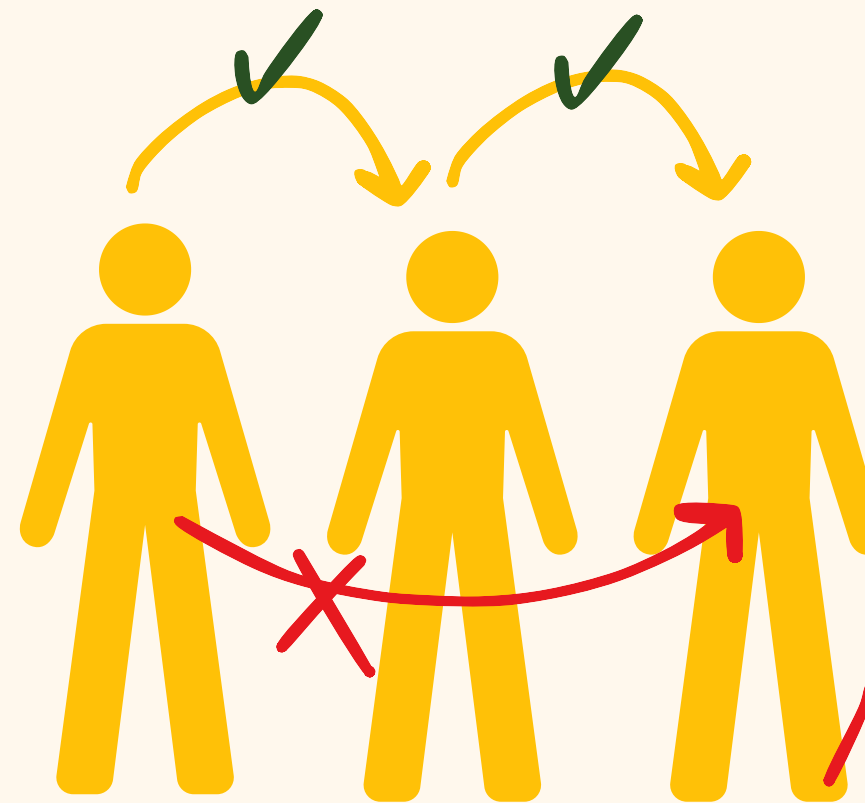
Game Demonstration (Group of 5)

Input: **GAGCTAAGC**
Output: **CUCGAUUCG**

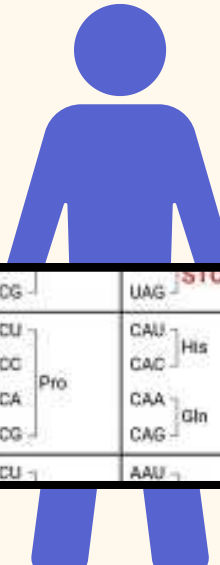
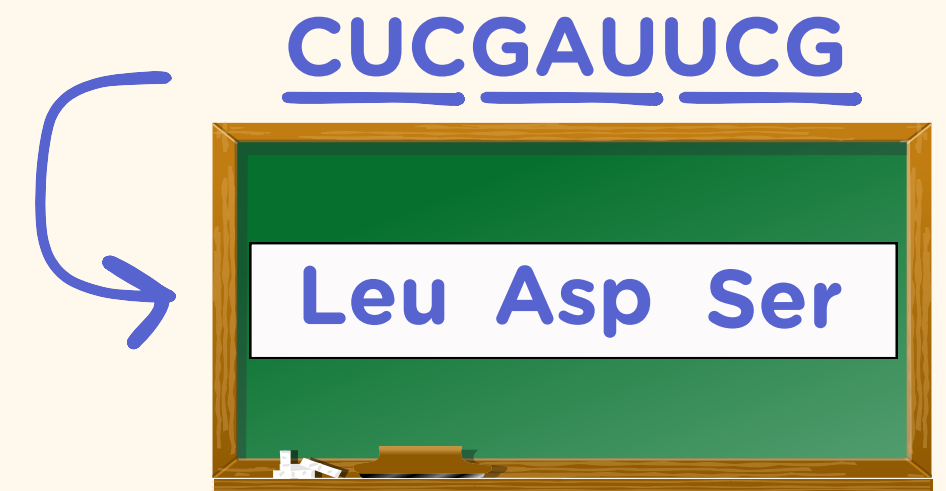


Sighter

CUCGAUUCG



Messenger



base in codon	UUG } Leu	UCG }	UAG } STOP	UGG } Trp	G	Last base in codon			
	C } CUU } Leu	CCU } Pro	CAU } His	CGU } Arg	U				
			CUC }		CCC }		CGC }	C	
			CUA }		CCA }		CAA } Gln	CGA }	A
			CUG }		CCG }		CAG }	CGG }	G
AUU }	ACU }	AAU }	AGU }	U					

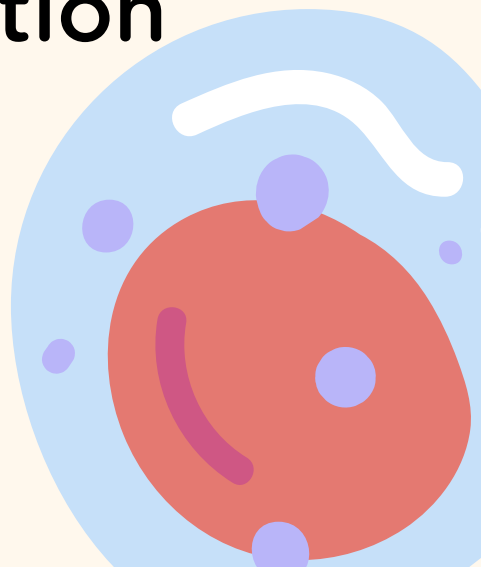
Builder

DNA-transcription and translation



Rules:

1. The sighter is not allowed to communicate with the builder.
2. Once you pass to the next person, you cannot talk with the person that pass it to you.
3. Each group has 4 minutes to finish passing the codon and done translation



Codon Chart

Second base in codon

		U	C	A	G		
First base in codon	U	UUU } Phe	UCU }	UAU } Tyr	UGU } Cys	U	Last base in codon
		UUC }	UCC } Ser	UAC }	UGC }	C	
		UUA } Leu	UCA }	UAA } STOP	UGA STOP	A	
		UUG }	UCG }	UAG }	UGG Trp	G	
	C	CUU }	CCU }	CAU } His	CGU }	U	
		CUC } Leu	CCC } Pro	CAC }	CGC } Arg	C	
		CUA }	CCA }	CAA } Gln	CGA }	A	
		CUG }	CCG }	CAG }	CGG }	G	
	A	AUU }	ACU }	AAU } Asn	AGU } Ser	U	
		AUC } Ile	ACC } Thr	AAC }	AGC }	C	
		AUA }	ACA }	AAA } Lys	AGA } Arg	A	
		AUG Met (start)	ACG }	AAG }	AGG }	G	
	G	GUU }	GCU }	GAU } Asp	GGU }	U	
		GUC } Val	GCC } Ala	GAC }	GGC } Gly	C	
		GUA }	GCA }	GAA } Glu	GGA }	A	
		GUG }	GCG }	GAG }	GGG }	G	

Answer:

Group 1 Met – Arg – Phe

Group 2 Asp – Lys – Gly

Group 3 Pro – Ser – Gln

Group 4 Cys – Asn – Gly

Group 5 Lys – Leu – Ser

Group 6 Val – Thr – [STOP]

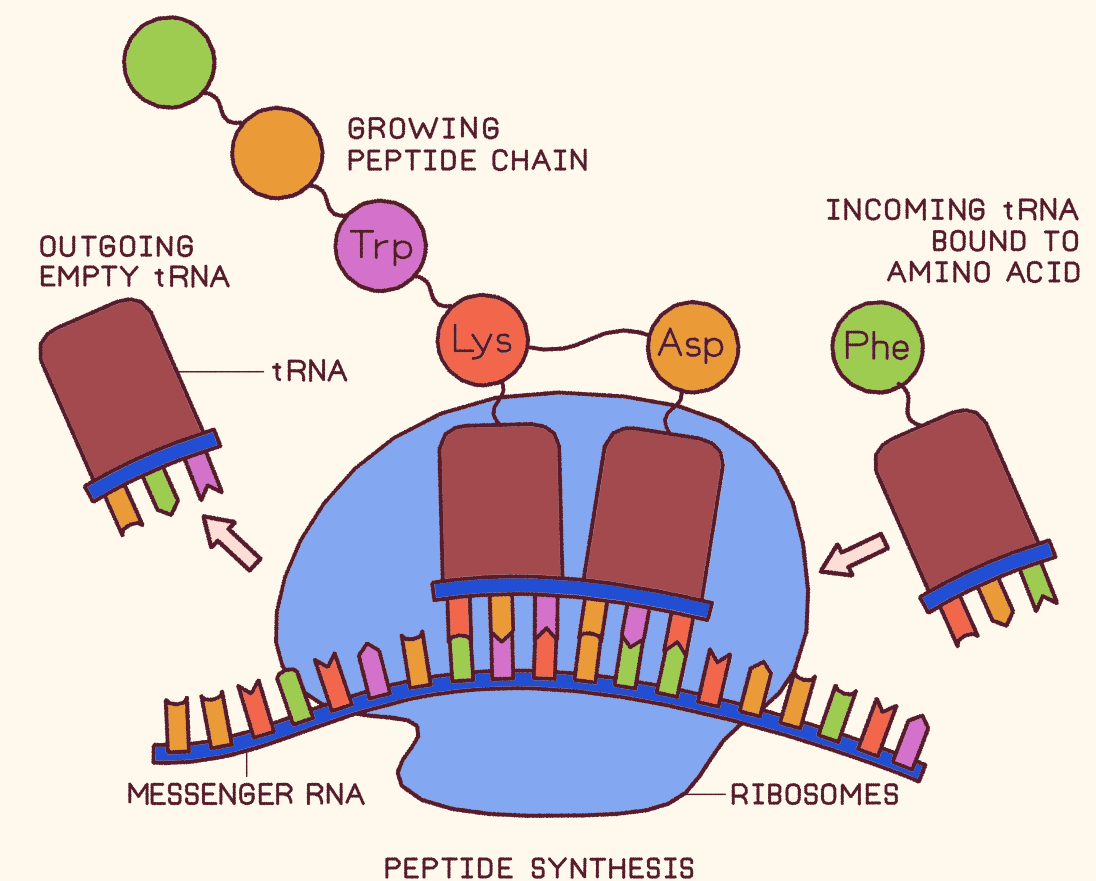
Codon Chart

		Second base in codon					
		U	C	A	G		
First base in codon	U	UUU } Phe UUC } UUA } Leu UUG }	UCU } UCC } Ser UCA } UCG }	UAU } Tyr UAC } UAA } STOP UAG }	UGU } Cys UGC } UGA } STOP UGG } Trp	U C A G	Last base in codon
	C	CUU } CUC } Leu CUA } CUG }	CCU } CCC } Pro CCA } CCG }	CAU } His CAC } CAA } Gln CAG }	CGU } CGC } Arg CGA } CGG }	U C A G	
	A	AUU } AUC } Ile AUA } AUG } Met (start)	ACU } ACC } Thr ACA } ACG }	AAU } Asn AAC } AAA } Lys AAG }	AGU } Ser AGC } AGA } Arg AGG }	U C A G	
	G	GUU } GUC } Val GUA } GUG }	GCU } GCC } Ala GCA } GCG }	GAU } Asp GAC } GAA } Glu GAG }	GGU } GGC } Gly GGA } GGG }	U C A G	

If you got the answer correctly...:

Leu Asp Ser

- This represents Normal Protein Synthesis
- The genetic information was transferred and translated without errors. The protein will fold correctly and perform its job

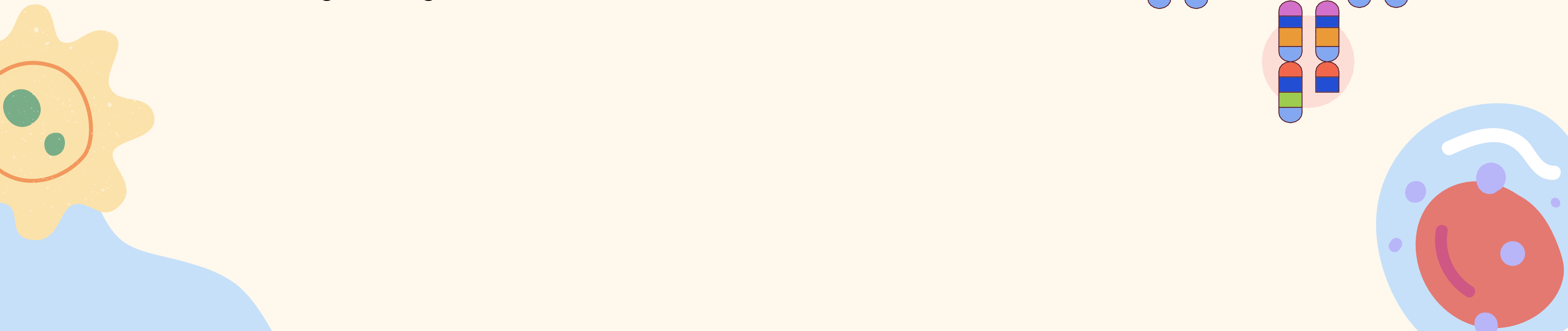
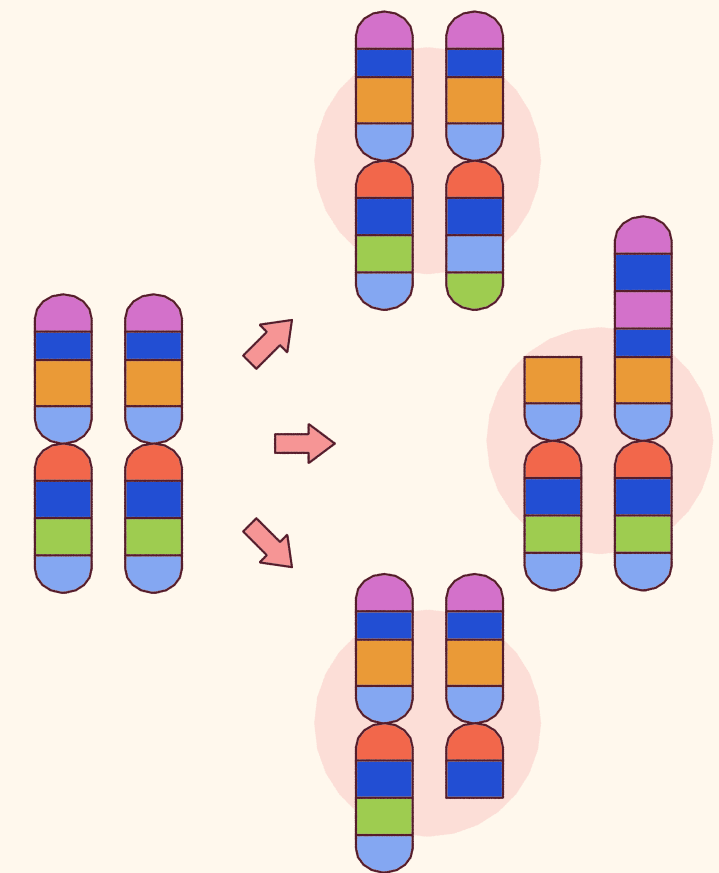


If you got the answer incorrect...:

- If the codon is not correct

→ **Mutation**

- Why do you think the mutation occurs?

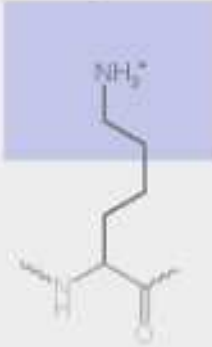
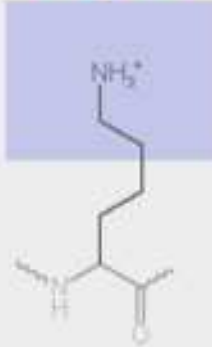
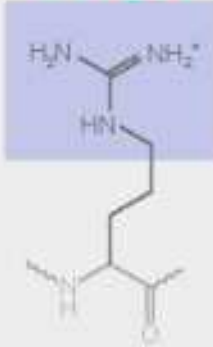
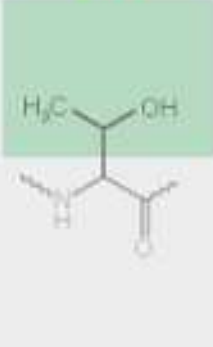




If you got the answer incorrect...:

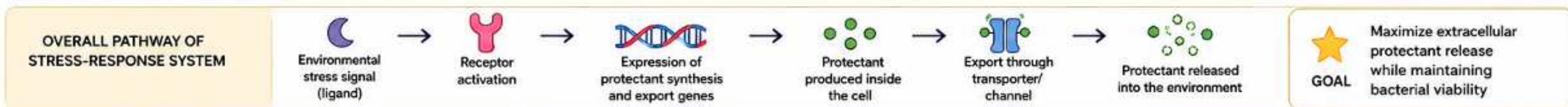
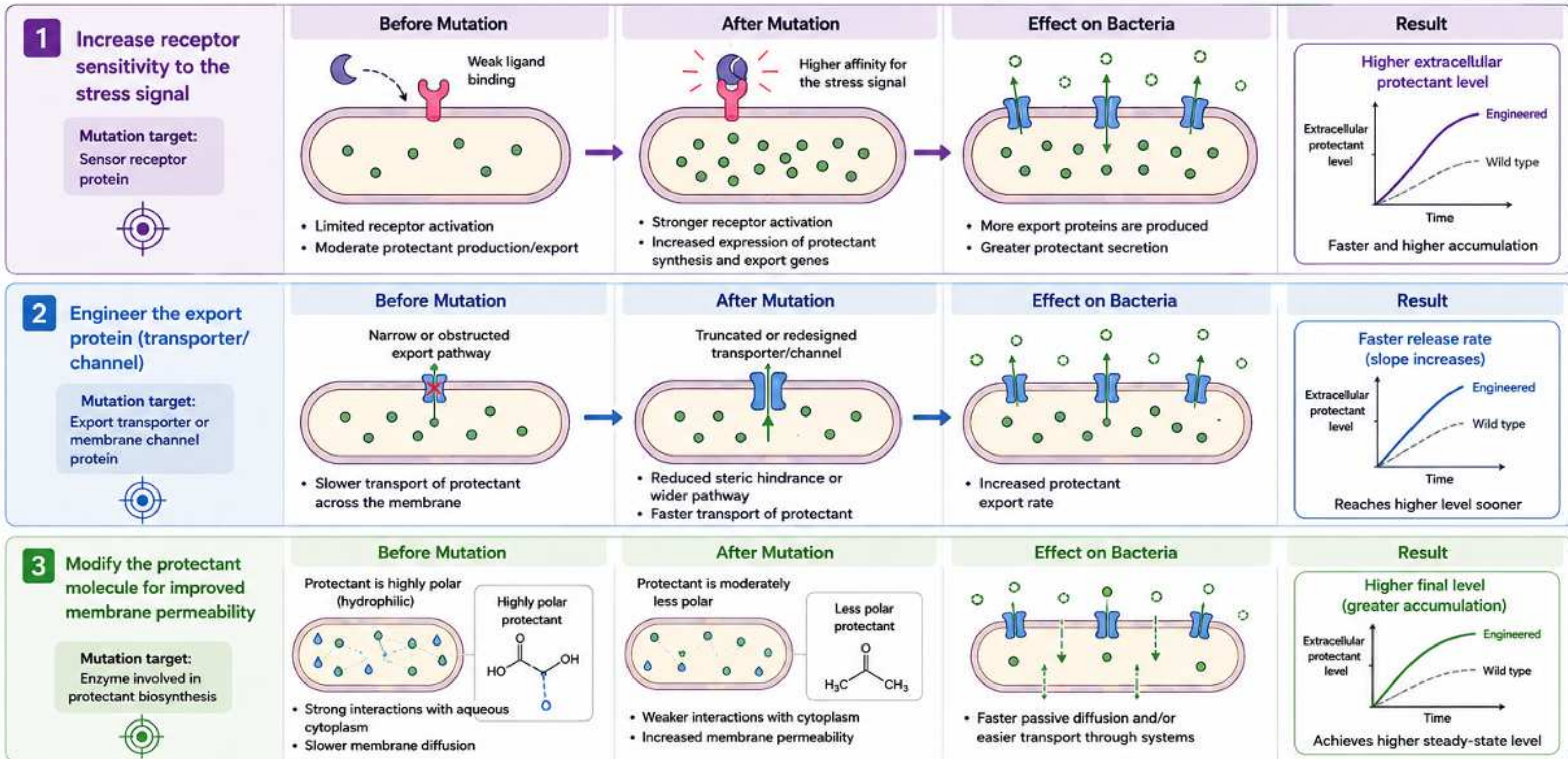
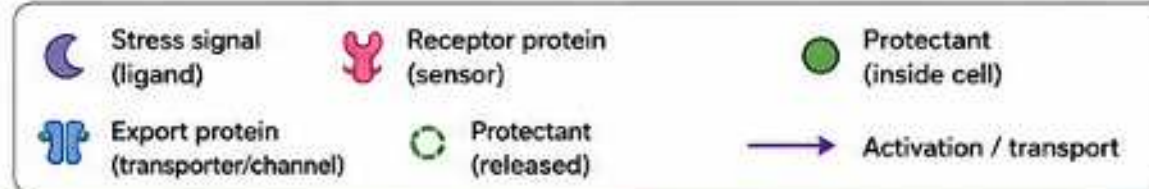
MUTATION (Fun fact: The Main Cause of Cancer)

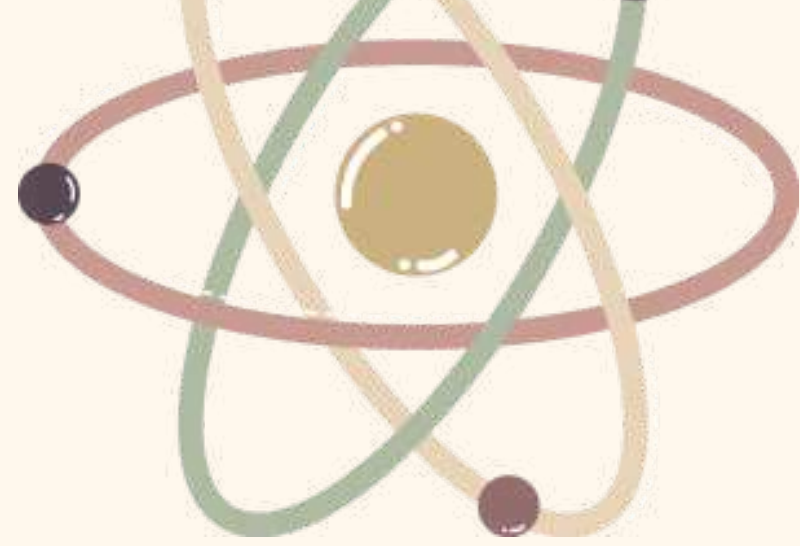
Normal DNA	TAT	CAT	CCT	AAG	GTA	
	└┐	└┐	└┐	└┐	└┐	
	↓	↓	↓	↓	↓	
Protein	Tyr	His	Pro	Lys	Val	
Substitution	TAT	CAT	CGT	AAG	GTA	
	└┐	└┐	└┐	└┐	└┐	
	↓	↓	↓	↓	↓	
Protein	Tyr	His	Arg	Lys	Val	
Insertion	TAT	CAT	CGC	TAA	GGT	A
	└┐	└┐	└┐	└┐	└┐	
	↓	↓	↓	↓	↓	
Protein	Tyr	His	Arg	Stop	Gly	
Deletion	TAT	C_TC	CTA	AGG	TA	
	└┐	└┐	└┐	└┐	└┐	
	↓	↓	↓	↓	↓	
Protein	Tyr	Leu	Leu	Arg	...	

	No mutation	Point mutations		
		Silent	Nonsense	Missense
				conservative non-conservative
DNA level	TTC	TTT	ATC	TCC TGC
mRNA level	AAG	AAA	UAG	AGG ACG
protein level	Lys	Lys	STOP	Arg Thr
				 
	basic	basic		basic polar

ENGINEERED MUTATIONS TO ENHANCE PROTECTANT RELEASE

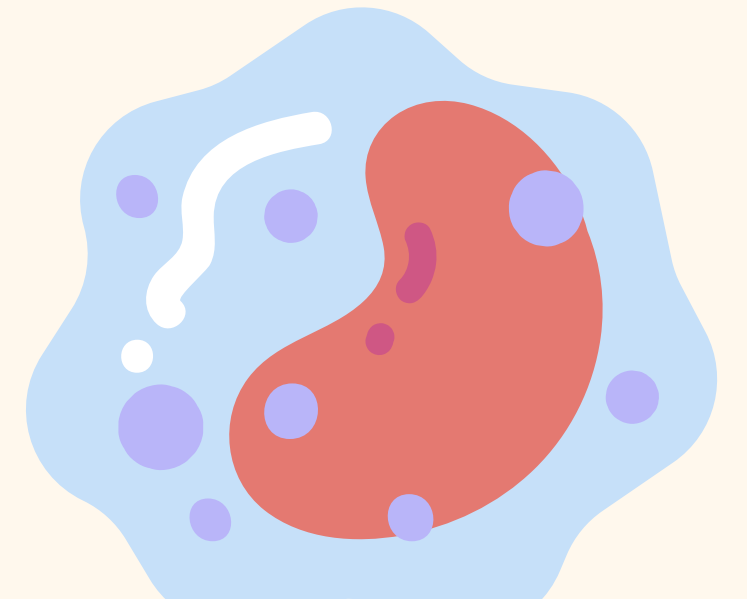
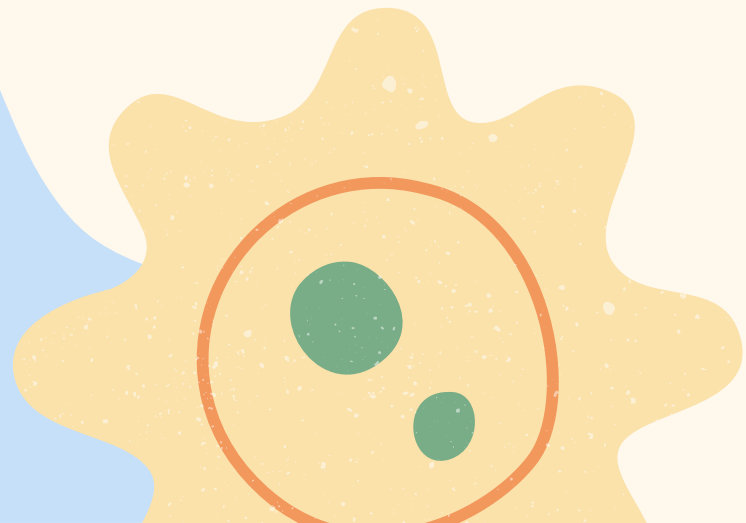
Different mutations target different components of the stress-response system to boost protectant release.





05

Course Summary



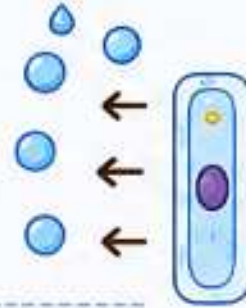
LESSON SUMMARY

Understanding Osmosis to Protect Plants
and Build a Better Future

KEY CONCEPTS

• Osmosis

Water moves from an area of higher water potential (lower solute concentration) to an area of lower water potential (higher solute concentration).



• Cells change in different concentration environments.

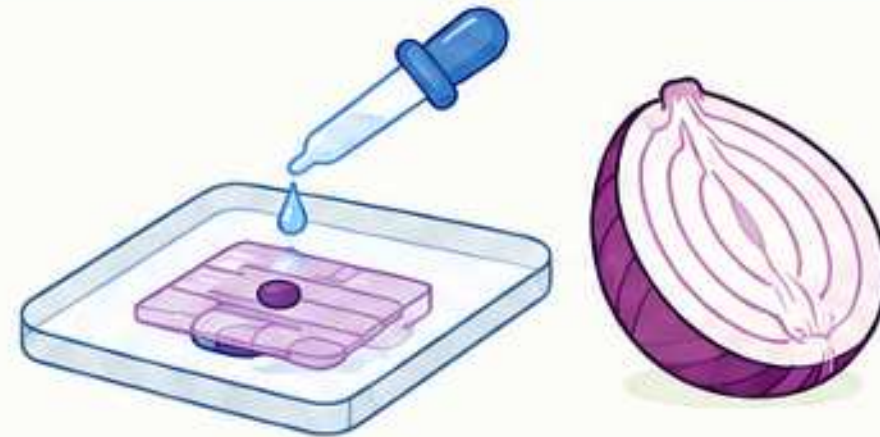
The direction of water movement affects cell size and turgor pressure.



• Solution concentration affects the movement of water in and out of cells.



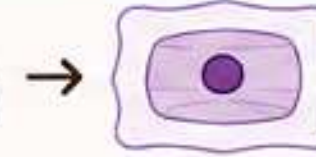
EXPLORING OSMOSIS IN ONION EPIDERMIS



By placing onion epidermis in solutions with different concentrations, we can observe how cells respond to changes in their environment.

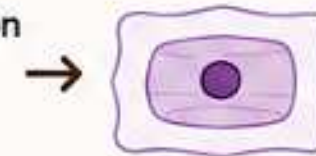
OBSERVATIONS

Low concentration
(Hypotonic solution)



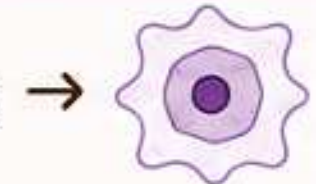
Cell swells and becomes turgid.

Isotonic concentration
(Similar to the cell interior)



No obvious change in the cell.

High concentration
(Hypertonic solution)



Cell shrinks and the membrane pulls away from the cell wall.

WHAT WE LEARNED



Water molecules move.

Water moves in or out of cells.

This affects plant cell growth and function.

Understanding osmosis helps us understand how plants survive and maintain life in different environments!



PLANT STRESS



When plants face drought, heat, or high salinity, water can leave cells, causing loss of turgor pressure.

This can lead to wilting, reduced growth, and lower crop yields.



SYNTHETIC BIOLOGY APPLICATIONS

Scientists use synthetic biology and genetic engineering to design crops that can better handle stress by:

- ✓ Improving water regulation
- ✓ Enhancing stress-response pathways
- ✓ Maintaining osmotic balance in cells



Goal: Develop crops that survive in dry or salty environments and secure future food supply.



MUTATION & ADAPTATION

Mutations are changes in DNA that can affect proteins involved in water transport or stress responses.

- ✗ Some mutations may reduce survival.
- ✓ Other mutations may increase tolerance to stress and become beneficial traits for crop improvement.

FUTURE CONNECTIONS & FOOD SECURITY



Apply knowledge of osmosis, genetics, and biotechnology to create climate-resilient crops.



Support sustainable agriculture and global food security for future generations.

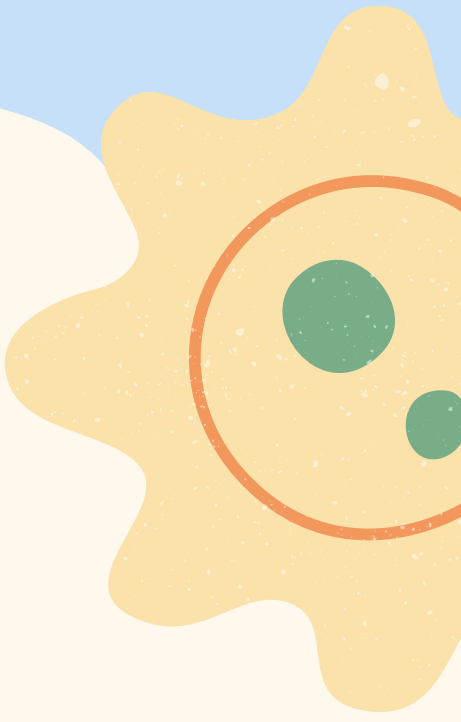
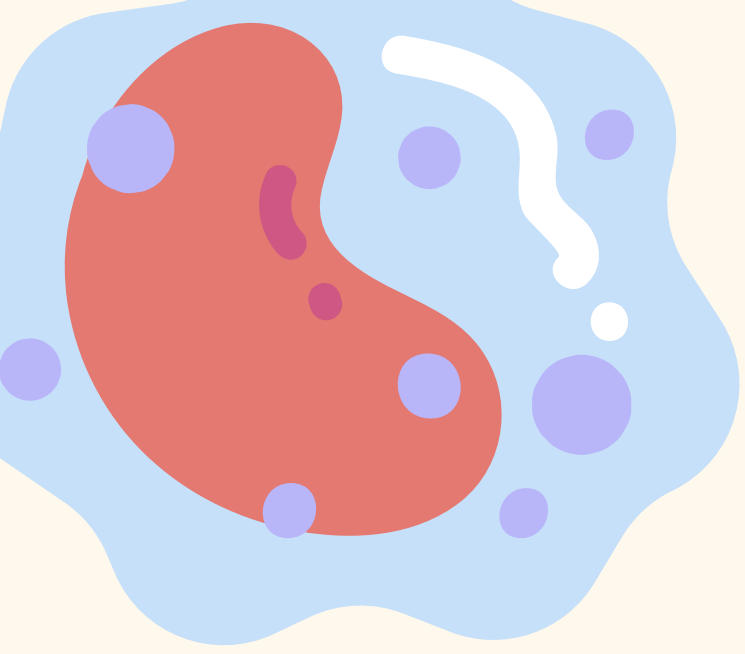


Science today, a better future tomorrow!



BIG IDEA: Osmosis connects cell biology, genetics, and biotechnology. By understanding how cells respond to their environment, we can use science and innovation to protect plants, adapt to climate change, and build a more food-secure future.





Thank You

